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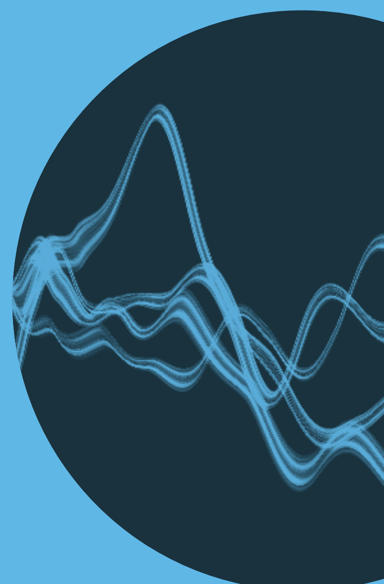
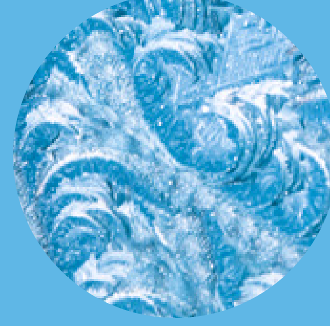
ERICSSON TECHNOLOGY Review



5G NEW RADIO EVOLUTION

NEXT-GENERATION **EDGE-CLOUD** ECOSYSTEM



PRIVACY-AWARE MACHINE LEARNING

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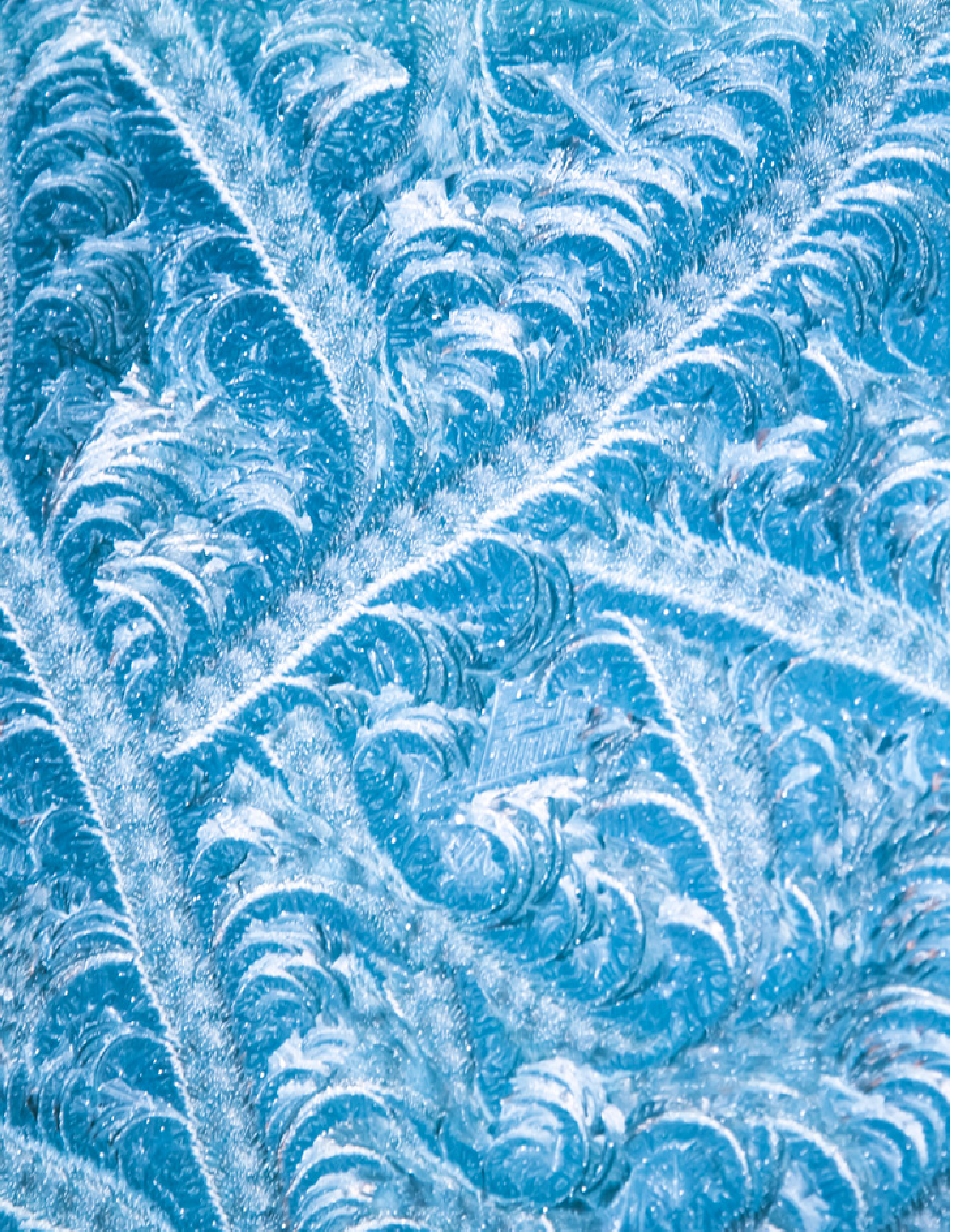

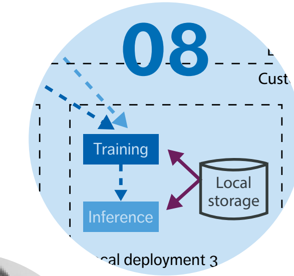
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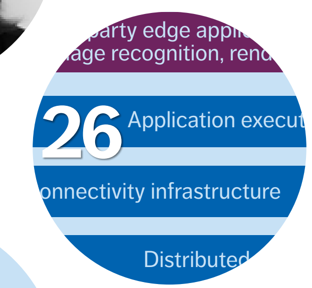
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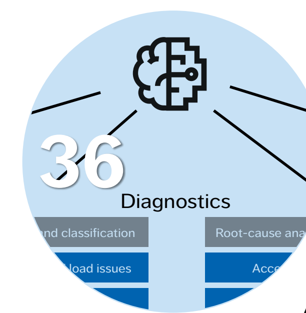
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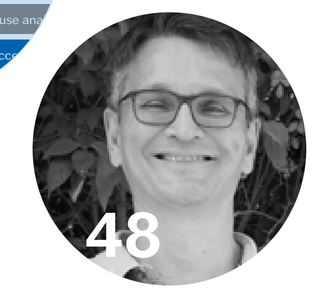
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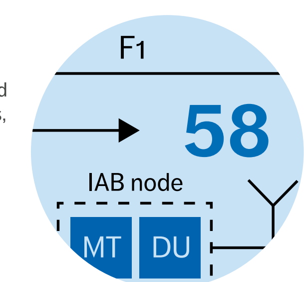
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Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities, and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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ADDRESSING 5G CHALLENGES TOGETHER

MOBILE DATA traffic volumes are expected to increase by a factor of four by 2025, and 45 percent of that traffic will be carried by 5G networks. To deliver on customer expectations in this rapidly changing environment, communication service providers (CSPs) must overcome challenges in three key areas: building sufficient capacity, resolving operational inefficiencies through automation and artificial intelligence (AI), and improving service differentiation. Fortunately, the contents of this issue of ETR magazine provide insights about how to tackle all three.

For many operators, the introduction of the 5G System (5GS) to provide wide-area services in existing Evolved Packet System (EPS) deployments is a critical step toward creating a full-service, future-proof 5GS in the longer term. Our article on the topic provides an overview of all the aspects that operators need to consider when putting together a robust EPS-to-5GS migration strategy and offers guidance on how to adapt the transition to address a CSP's specific needs per domain.

To cope with the large increase in required bit rate per site and achieve a cost-efficient rollout of 5G New Radio (NR), CSPs also need a good understanding of the different RAN architecture and transport network alternatives available to them. In this issue, we present all the available options and explain why automation and tight integration between the RAN and the transport network are critical to achieving cost-efficient deployments.

The surge in data volume that will come from the massive number of devices enabled by 5G has made edge computing more important than ever before. Beyond its abilities to reduce network traffic and improve user experience, edge computing will also play a critical role in enabling use cases for ultra-reliable low-latency communication in industrial manufacturing and a variety of other sectors. Our article on the topic explores how to deliver distributed edge computing solutions that can host different kinds of platforms and applications and provide a high level of flexibility for application developers.

> AUTOMATION AND TIGHT INTEGRATION... ARE CRITICAL TO ACHIEVING COST-EFFICIENT DEPLOYMENTS

The integration of AI into current and future generations of cellular access will be critical to achieving Ericsson's vision of creating a cellular network that constantly adapts itself both to customer requirements and to the static and dynamic characteristics of different scenarios. The AI article in this issue explains how AI can be applied most effectively in three RAN performance improvement domains: network design, network optimization and RAN algorithms.

This issue of the magazine also features an article about federated learning (FL) – a smarter, more resource-efficient way for CSPs to ensure consistent QoE. The article demonstrates that it is possible to migrate from a conventional machine learning model to an FL model and significantly reduce the amount of information that is exchanged between different parts of the network, thereby enhancing privacy without negatively impacting accuracy.

Ericsson is deeply committed to helping CSPs and other stakeholders understand and plan for the many new 5G NR opportunities that are on the horizon. The significant enhancements to 5G NR in 3GPP releases 16 and 17 are certain to play a critical role in expanding both the availability and the applicability of 5G NR in both industry and public services. Our article on this topic analyzes the most notable new developments in these coming releases, and shares our insights about the future beyond release 17.

We hope you enjoy this issue of ETR magazine and we'd be delighted if you shared it with your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review

ERIK EKUDDEN
SENIOR VICE PRESIDENT,
CHIEF TECHNOLOGY OFFICER AND
HEAD OF GROUP FUNCTION TECHNOLOGY

Privacy-aware machine learning

WITH LOW NETWORK FOOTPRINT

Federated learning makes it possible to train machine learning models without transferring potentially sensitive user data from devices or local deployments to a central server. As such, it addresses privacy concerns at the same time that it improves functionality. Depending on the complexity of the neural network, it can also dramatically reduce the amount of data needed while training a model.

KONSTANTINOS VANDIKAS, SELIM ICKIN, GAURAV DIXIT, MICHAEL BUISMAN, JONAS ÅKESON

Reliance on artificial intelligence (AI) and automation solutions is growing rapidly in the telecom industry as network complexity continues to expand. The machine learning (ML) models that many mobile network operators (MNOs) use to predict and solve issues before they affect user QoE are just one example.

An important aspect of the 5G evolution is the transformation of engineered networks into continuous learning networks in which self-adapting, scalable and intelligent agents can work independently to continuously improve quality and performance. These emerging "zero-touch networks" are, and will continue to be, heavily dependent on ML models.

The real-world performance of any ML model depends on the relevance of the data used to train it. Conventional ML models depend on the mass transfer of data from the devices or deployment sites to a central server to create a large, centralized dataset. Even in cases where the computation is decentralized, the training of conventional ML models still requires large, centralized datasets and misses out on using computational resources that may be available closer to where data is generated.

While conventional ML delivers a high level of accuracy, it can be problematic from a data security perspective, due to legal restrictions and/or privacy concerns. Further, the transfer of so much data requires significant network resources, which means that lack of bandwidth and data transfer costs can be an issue in some situations. Even in cases where all the required data is available, reliance on a centralized dataset for maintenance and retraining purposes can be costly and time consuming.

For both privacy and efficiency reasons, Ericsson believes that the zero-touch networks of the future must be able to learn without needing to transfer voluminous amounts of data, perform centralized computation and/or risk exposing sensitive information. Federated learning (FL), with its ability to do ML in a decentralized manner, is a promising approach.

To better understand the potential of FL in a telecom environment, we have tested it in a number of use cases, migrating the models from conventional, centralized ML to FL, using the accuracy of the original model as a baseline. Our research indicates that the usage of a simple neural network yields a significant reduction in network utilization, due to the sharp drop in the amount of data that needs to be shared.

As part of our work, we have also identified the properties necessary to create an FL framework that can achieve the high scalability and fault tolerance required to sustain several FL tasks in parallel. Another important aspect of our work in this area has been figuring out how to transfer an ML model that addresses a specific and common problem, pretrained within an FL mechanism on existing network nodes to newly joined network nodes, so that they too can benefit from what has been learned previously.

The concept of federated learning

The core concept behind FL is to train a centralized model on decentralized data that never leaves the local data center that generated it. Rather than transferring "the data to the computation," FL transfers "the computation to the data."[1]

In its simplest form, an FL framework makes use of neural networks, trained locally as close as possible to where the data is generated/collected. Such initial models are distributed to several data sources and trained in parallel. Once trained, the weights of all neurons of the neural network are transported to a central data center, where federated averaging takes place and a new model is produced and communicated back to all the remote neural networks that contributed to its creation.

Terms and abbreviations

AI – Artificial Intelligence | **AUC** – Area Under the Curve | **FL** – Federated Learning | **ML** – Machine Learning | **MNO** – Mobile Network Operator | **ROC** – Receiver Operating Characteristic

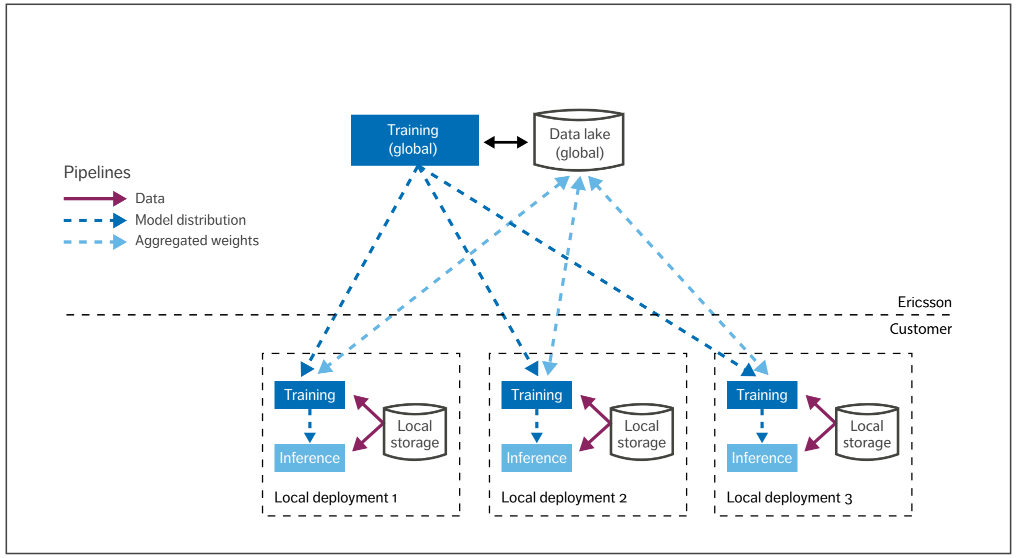

Figure 1 Overview of federated learning

Techniques such as secure aggregation [2] and differential privacy [3] can be applied to further ensure the privacy and anonymity of the data origin.

FL can be used to test and train not only on smartphones and tablets, but on all types of devices. This makes it possible for self-driving cars to train on aggregated real-world driver behavior, for example, and hospitals to improve diagnostics without breaching the privacy of their patients.

Figure 1 illustrates the basic architecture of an FL life cycle. The light blue dashed lines indicate that only the aggregated weights are sent to the global data lake, as opposed to the local data itself, as is the case in conventional ML models. As a result, FL makes it possible to achieve better utilization of resources, minimize data transfer and preserve the privacy of those whose information is being exchanged.

The main challenge with an FL approach is that the transition from training a conventional ML model using a centralized dataset to several smaller federated ones may introduce a bias that impacts the accuracy originally achieved by using a centralized dataset. The risk for this is greatest in less reliable and more ephemeral federations that span over to mobile devices.

It is reasonable to expect data centers used by MNOs to be significantly more reliable than devices in terms of data storage, computational resources and general availability. However, it is important to ensure high fault tolerance, as corresponding processes may still fail due to lack of resources, software bugs or other issues.

Federated learning framework design

Our FL framework design concept is cloud-native, built on a federation of Kubernetes-based data centers located in different parts of the world. We assume restricted access to allow for the execution of certain processes that are vital to FL.

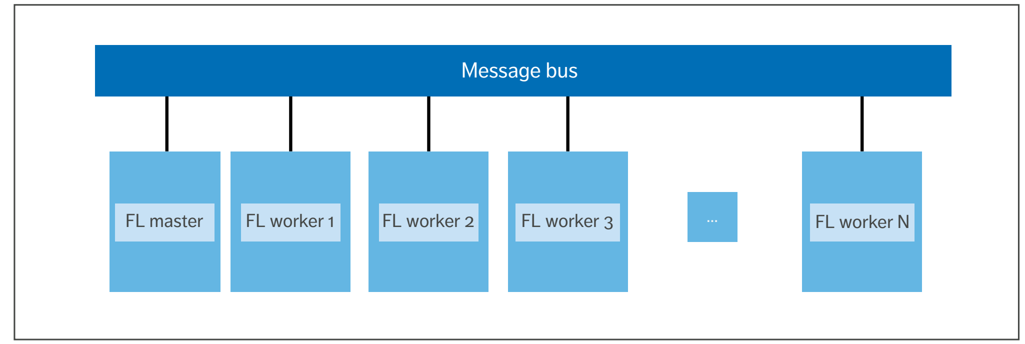

Figure 2 Basic design of an FL platform

Figure 2 illustrates the basic system design. A federation is treated as a task run-to-completion, enabling a single resource definition of all parameters of the federation that is later deployed to different cloud-native environments. The resource definition for the task deals both with variant and invariant parts of the federation.

The variant parts handle the characteristics of the FL model and its hyperparameters. The invariant parts handle the specifics of common components that can be reused by different FL tasks. Invariant parts include a message queue, the master of the FL task and the workers to be deployed and federated in different data centers.

Workers (processes running in local deployments) are tightly coupled with the underlying ML platform that is used to train the model, which is immutable during the federation. In our FL experiments, we selected TensorFlow to train the neural network, which is designed to be interchangeable with other ML platforms such as PyTorch. Communication between the master and the workers is protected using Transport Layer Security encryption with one-time generated public/private keys that are discarded as soon as an FL task is completed.

Invariant components can be reused by different FL tasks. FL tasks can run sequentially or in parallel depending on the availability of resources. Master and worker processes are implemented as stateless components. This design choice leads to a more robust framework, since it allows for an FL task to fail without affecting other ongoing FL tasks.

Fault tolerance

To reduce the complexity of the codebase for both the master of the FL task and the workers and to keep our implementation stateless, we chose a message bus to be the single point of failure in the design of our FL framework. This design choice is further motivated by the research into creating highly scalable and fault-tolerant message buses by combining leader-election techniques and/or by relying on persistent storage to maintain the state of the message queue in case of a failure [4].

The message exchange between the master of the FL task and the workers is implemented in the form of assigned tasks such as "compute new weights" and "average weights." Each task is pushed into the message queue and has a direct recipient. The recipient must acknowledge that it has received the task. If the acknowledgement is not made, the task remains in the queue. In case of a failure, messages remain in the message queue while Kubernetes restarts the failed process. Once the process reaches a running state again, the message queue retransmits any unacknowledged tasks.

Table A – Summary of scenario definitions

	Centralized	Isolated	Federated
Privacy preserved	No	Yes	Yes
Use of overall data	Yes	No	Yes
Data transfer cost	High	None	None
Weight transfer cost	None	None	Low

Table B – ROC AUC scores of workers throughout three scenarios

	Centralized median (std)	Isolated median (std)	Federated median (std)
Worker 1 (region 1)	0.91 (0.15)	0.89 (0.12)	0.95 (0.05)
Worker 2 (region 2)	0.92 (0.8)	0.93 (0.08)	0.93 (0.03)
Worker 3 (region 3)	0.95 (0.16)	0.95 (0.13)	0.97 (0.07)
Worker 4 (region 4)	0.97 (0.13)	0.97 (0.11)	0.96 (0.05)
Overall	0.93 (0.13)	0.93 (0.11)	0.95 (0.05)

Table C – Network footprint formulas for each training scenario

		Downlink consumption	Uplink consumption
Centralized ML	Master	$\sum_{i=0}^{N} n_i$	$N * \text{Model}_0$
	Worker ID	Model_0	n_i
Isolated ML	Master	0	0
	Worker ID	0	0
FL	Master	$N * R * \text{Model}_0$	$N * R * \text{Model}_0$
	Worker ID	$R * \text{Model}_0$	$R * \text{Model}_0$

i: worker ID
N: number of workers
R: number of rounds needed until accuracy convergence
Model_0: Size of ML model
n_i: size of dataset in worker ID

Table D – Network footprint

Centralized (MB)	Federated (MB)	FL message size (MB)	Rounds	Workers
2,000	19.2	0.26	15	4

Figure 3 Tables relating to the hardware fault prediction use case

Preventive maintenance use case

Hardware fault prediction is a typical ML use case for an MNO. In this case, the aim is to predict whether there will be a hardware fault at a radio unit within the next seven days based on data generated in the eight-week interval preceding the prediction time. The inputs to the ML model consist of more than 500 features that are aggregations of multiple performance management counters, fault management data such as alarms, weather data and the date/time since the hardware has been active in the field.

Three training scenarios

We performed the experiments in three scenarios – centralized ML, isolated ML and FL.

Centralized ML is the benchmark scenario. The datasets from all four worker nodes are transferred to one master node, and model training is performed there. The trained model is then transferred and deployed back to the four worker nodes for inference. In this scenario, all worker nodes use exactly the same pretrained ML model.

In the isolated ML scenario, no data is transferred from the worker nodes to a master node. Instead, each worker node trains on its own dataset and operates independently from the others.

In the FL scenario, the worker nodes train on their individual datasets and share the learned weights from the neural network model via the message queue. The saturation of the model accuracies is achieved after 15 rounds of the weight-sharing and weight-averaging procedure. In this way, the worker nodes can learn from each other without transferring their datasets.

The properties of each training scenario are summarized in *Figure 3*, Table A.

Accuracy results

Table B in Figure 3 presents the results in the form of median ROC AUC (receiver operating characteristic area under the curve) scores obtained through more than 100 independent experiments. The scores achieved in the FL scenario are similar to those achieved in the centralized and isolated ones, while the variance of the FL scores is significantly lower compared with the other two scenarios.

The results in Table B show that it is worker 1 (south) that benefits from FL. They also suggest that an isolated ML approach can be recommended in cases where the individual datasets have enough data for training. The only drawback is that because the isolated nodes never receive any information from other nodes, they will be more conservative in their response to changes in the data, with the risk of potentially higher blind spots in the individual datasets.

The impact of adding new workers

To facilitate the adding of new workers at a later time, information about the current round must be maintained in the message exchange between the master and the workers. When an FL task starts, all workers register to round ID 0, which triggers the master to initialize the random weights and broadcast the same distribution to all workers. All workers train in parallel and contribute to the same training round. As the rounds increase, the federated model's maturity increases until a saturation point is reached.

If the current round ID is greater than 0, the master is aware that the process of averaging of weights has taken place at least once, which means that the model is not at a random initial state. When a new worker joins the FL task, it sends its round ID as 0.

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5G New Radio RAN & transport

CHOICES THAT MINIMIZE TCO

By deploying self-built transport in the RAN area instead of using leased lines, mobile network operators gain access to the full range of 5G New Radio RAN architecture options and minimize their total cost of ownership (TCO).

ANN-CHRISTINE ERIKSSON,
MATS FORSMAN,
HENRIK RONKAINEN,
PER WILLARS,
CHRISTER ÖSTBERG

The 5G evolution is well underway – leading mobile network operators (MNOs) in several regions of the world have already launched the first commercial 5G NR networks, and large-scale deployments are expected in the years ahead. The use of self-built transport in denser areas with a suitable RAN architecture will play a key role in ensuring cost-efficiency.

A cost-efficient 5G NR deployment requires MNOs to take several factors into consideration. Most obviously, they need to make sure that the 5G NR deployment complements their existing 4G LTE network and makes use of both current 4G LTE and new 5G NR spectrum assets. Beyond that, it is vital to consider the various RAN architecture options available and the ways in which the transport network needs to evolve to support them, along with the large increase in user data rates per site.

While urban areas with high user density will be the first priority for 5G NR deployments, suburban and rural areas will not be far behind. These three area types have different preconditions such as available transport solutions, inter-site distance (ISD), traffic demand and spectrum needs that must be taken into consideration at an early stage in the deployment process.

Predicted 5G traffic

5G is projected to reach 40 percent population coverage and 1.9 billion subscriptions by 2024 [1], corresponding to 20 percent of all mobile subscriptions. Those figures indicate that it will be the fastest global rollout so far. The total mobile data traffic generated by smartphones is currently about 90 percent and is estimated to reach 95 percent by the end of 2024. With the continued growth of smartphone usage, total worldwide mobile data traffic is predicted to reach about 130 exabytes per month – four times higher than the corresponding figure for 2019 – and 35 percent of this traffic will be carried by 5G NR networks.

The growing data demands for mobile broadband can generally be met with limited site densification [2]. There are benefits to deploying 5G NR mid-bands (3-6GHz) at existing 4G sites, resulting in a significant performance boost and maximal reuse of site infrastructure investments. By means of massive MIMO (multiple-input, multiple-output) techniques, such as beamforming and multi-user MIMO, higher downlink capacity can be achieved along with improved downlink data rates – both outdoors and indoors.

Deep indoor coverage is maintained through interworking with LTE and/or NR on low bands using dual connectivity or carrier aggregation. Further speed and capacity increases can be attained by deploying 5G NR at high bands (26-40GHz), also known as mmWave. If additional spectrum does not satisfy the traffic demand (due to, for example, the introduction of fixed wireless access) densification with solutions such as street sites may be required.

Increasing user data rates per antenna site

The introduction of new spectrum for 5G NR will increase the carrier bandwidths from the 5MHz, 10MHz and 20MHz used for LTE to 50MHz and 100MHz for the mid bands (3-6GHz) and 400/800MHz for the high bands (24-40GHz), allowing for gigabit-per-second data rates per user equipment (UE). In urban areas, the total amount of spectrum will grow from a few tens or hundreds of megahertz to several hundred or thousand megahertz per antenna site.

Simultaneously, traffic demands per subscriber will increase exponentially. All in all, this implies that the bitrate demands in the backhaul and fronthaul transport network will increase significantly (per antenna site, for example). The bitrate demand will be multiple gigabits per second, compared with the few hundred megabits per second in current mobile networks.

The spectrum increase per antenna site will be less in suburban areas, while in rural areas refarming of current spectrum or spectrum sharing between LTE and NR will be more common. RAN transport networks will need to evolve to address the increase in accumulated user data rates, particularly in urban areas, and in many suburban ones as well.

Transport network options

Evolving the transport network in the local RAN area is an important first step when deploying 5G on top of LTE.

In most cases, the mobile backhaul transport for Distributed RAN (DRAN) – the architecture traditionally used to build mobile networks – has been a rented packet-forwarding service, Ethernet or IP based, typically called a leased line and provided by traditional fixed network operators. Another option is white fiber, an optical wavelength service offered by many traditional fixed network operators.

Instead of leasing a transport service, some mobile operators deploy self-built transport solutions using microwave links, which usually enables short installation lead time. Integrated Access and Backhaul (IAB) is another option for self-built transport in 5G. With IAB, the mobile spectrum is also used for backhaul, which is especially relevant for high-frequency bands where the bandwidth may be hundreds of megahertz.

Alternatively, it is possible for a mobile operator to deploy a self-built transport solution on top of physical fiber (known as dark fiber) that is available for rent from fixed network operators, or more recently from pure fiber network operators and municipal networks. The mobile operator then builds and owns the transport equipment in a RAN area, defined as the local urban area in a city and the suburban areas close to cities.

Urban areas tend to have multiple fiber network operators that deploy fiber to every street, which means that dark fiber is readily available for rent. While dark fiber is less common in suburban areas,

> TRAFFIC DEMANDS PER SUBSCRIBER WILL INCREASE EXPONENTIALLY

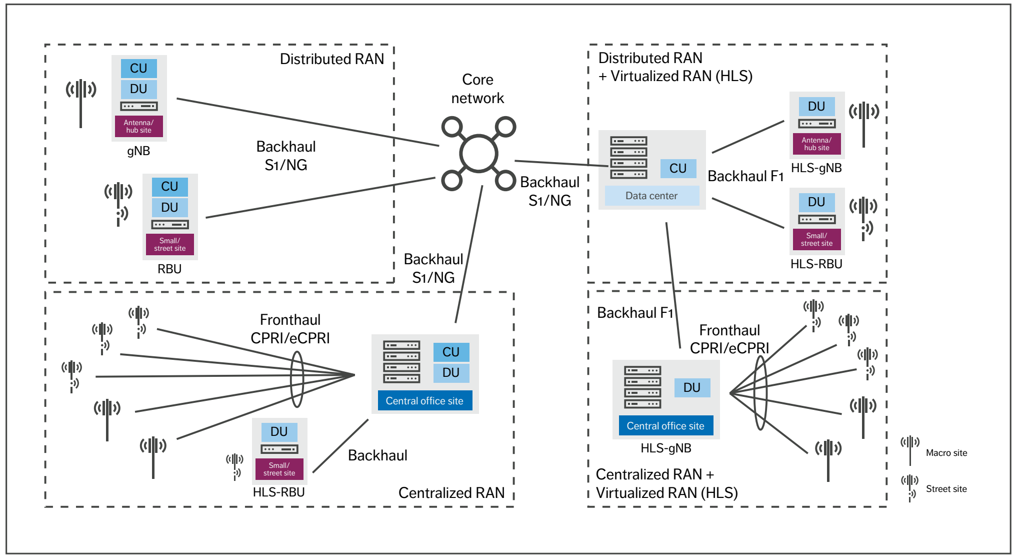

Figure 1 RAN architecture deployment options

its availability is steadily increasing. In rural areas, there is often only one fiber operator, and fiber is only deployed to specific sites such as businesses and schools. In these cases, dark fiber is usually not provided as a service.

On top of dark fiber, mobile operators can deploy an optical (passive or active) or a packet-forwarding solution. The passive optical solution uses colored small form-factor pluggable transceivers (SFPs) in the endpoints and optical filters in between for add/drop to subtended sites/equipment along the fiber path. An active optical system uses gray SFPs in the endpoints and active optical switching equipment to generate wavelengths and perform optical switching on the sites/equipment on the fiber. The packet-forwarding solution can be an Ethernet or IP solution with packet-forwarding capabilities on all sites/equipment along the fiber path.

RAN architecture options

Figure 1 illustrates DRAN along with the other RAN architecture options available for use in 5G NR. The option that is most appropriate for a particular deployment will largely depend on the type of deployment area (urban, suburban or rural) and the availability of dark fiber.

In all options, outdoor site deployments can be either macro sites (typically mounted on rooftops or antenna masts covering a larger area) or street sites (typically mounted on poles, walls or strands covering smaller areas or spots).

The flexibility of locating RAN functionality in different locations in 5G NR RAN architecture and the ability to support more radio sites increases the need for network automation, making it necessary to simplify the installation, deployment and operation of both the RAN and transport pieces. For example, the automation capabilities used to simplify installation in the RAN must also be introduced into transport to improve the interaction between the two.

Distributed RAN

DRAN with unitary eNodeB base stations has been the dominant architecture for 4G LTE. DRAN will also be a commonly used architecture in 5G NR deployments, with the benefit of reusing the legacy infrastructure investments. The backhaul – that is, the transport between the RAN and the core network (CN) – uses an S1/NG interface [3].

DRAN is well suited for use in all areas (urban, suburban and rural) and can use a large variety of transport solutions. DRAN reuses legacy infrastructure investments, such as existing sites and operations and maintenance structure, and is of particular value in areas where the population density is low and the users are scattered. The utilization of statistical variations in traffic for the dimensioning of self-built packet transport in the RAN area transport network is another benefit of DRAN.

Where densification is needed for coverage or capacity, DRAN street sites fit well together with the existing DRAN macro sites. Specific DRAN units tailored for street sites, denoted as RBU in Figure 1, have benefits such as integrated baseband functions, simple installation and reduced street site space.

> THE USE OF DARK FIBER IS A GOOD FIT WITH THE NEW WIDE NR FREQUENCY BANDS...

Centralized RAN

Centralized RAN (CRAN) is characterized by centralized baseband for multiple pieces of radio equipment. With a CRAN deployment, the baseband units located in a central site and the radio equipment located at the antenna sites are interconnected with a transport network denominated fronthaul, either Common Public Radio Interface (CPRI) or evolved CPRI (eCPRI) [4].

In areas with small ISDs and access to dark fiber (urban and in some cases dense suburban areas), centralizing and pooling the baseband units to an aggregation site can be a good option. The use of CRAN can lead to reduced costs for site space and energy consumption at the antenna sites, as well as easier installation, operation and maintenance.

CRAN provides efficient coordination (via interband carrier aggregation and CoMP – coordinated multipoint – for example) between physically separated antenna sites. It also enables dimensioning of a baseband pool to handle more and larger antenna sites due to statistical variations of traffic per site, which also makes baseband resource expansion easier when traffic grows in the CRAN area. Resilience and energy efficiency are other benefits, as the baseband pool serves many antenna sites. The statistical variation of traffic per site may also be utilized in RAN area transport network dimensioning.

In environments where CRAN is deployed, a dark fiber transport solution is required for the fronthaul. The connected radio sites also need to be within the latency limit required by the baseband units. The use of dark fiber is a good fit with the new wide NR frequency bands and the expansion of the fronthaul due to the use of advanced antenna systems [5].

When deploying CRAN, it is most beneficial to connect sites in the same area to the same baseband pool. In cases where it is difficult to deploy a dark fiber transport solution, either a DRAN or a high-layer split virtualized RAN (HLS-VRAN) architecture may be deployed for those sites, coexisting with other CRAN-connected nodes.

To achieve the benefits of statistical multiplexing of traffic to/from the radio equipment in the transport network and in the baseband pool, it is necessary to use an Ethernet-based fronthaul such as eCPRI [4]. The radio equipment at the antenna sites may either have support for eCPRI or include a converter from CPRI to eCPRI. It is also possible to mix eCPRI and CPRI radio equipment, using an optical fronthaul transport solution, but without transport multiplexing gains.

CRAN requires suitable sites (such as central office sites) to colocate the baseband units. The size and density of these central office sites depends on each situation, but a typical case could be central office sites with an ISD of less than 1km up to a few kilometers in an area.

Higher-layer split applied as a virtualized RAN deployment

For both DRAN and CRAN, it is possible to add a VRAN by implementing an HLS where the gNB

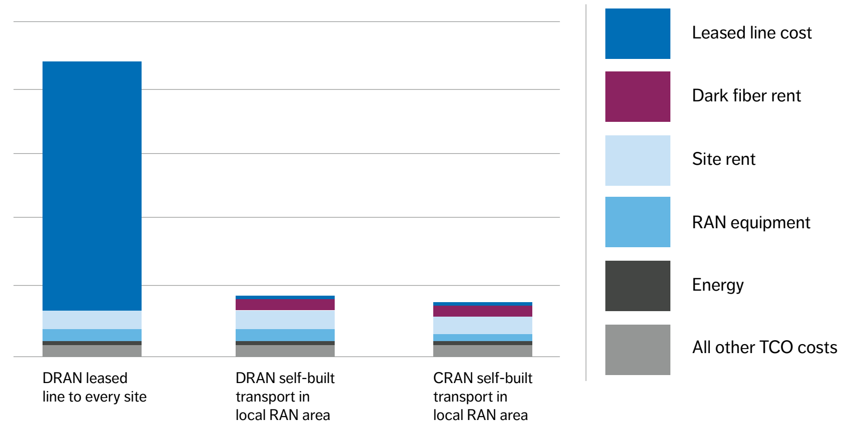

Figure 2 Relative operator TCO for 5G NR introduction in an urban local RAN area

is divided into a central unit (CU) and distributed units (DUs). This is known as HLS-VRAN. The DUs and the CU are separated by the F1 interface, carried on a backhaul transport network. These are denoted HLS-gNB for macro and HLS-RBU for street sites in Figure 1.

When a cloud infrastructure already exists in the network, the HLS-VRAN deployment may be beneficial from an operational and management point of view. For a DRAN deployment, adding HLS-VRAN could result in dual connectivity gains if it is expected that it will be common for UEs to be connected to different baseband sites.

In areas where a street site deployment is needed as a coverage or capacity complement to the macro site deployment, a street HLS-VRAN deployment fits well with macro HLS-VRAN. Specific HLS-VRAN units tailored for street sites, denoted as HLS-RBU in Figure 1, have the same benefits as the RBU.

5G New Radio total cost of ownership

A mobile operator's TCO for 5G NR introduction in a RAN area includes both capital expenses (one-time costs) and operating expenses (recurring costs). Typical capital expenses include radio/RAN and transport equipment, site construction, installation costs and site acquisition. Typical operating expenses include costs for a leased line, dark fiber rental, spectrum for wireless transport, site rental, energy consumption, operation and maintenance costs and vendor support. Since the RAN area type and deployment solution alternatives affect the TCO, it is useful to compare the TCO of the deployment solution alternatives in different RAN areas.

Based on Ericsson customer price information and internal analysis, *Figure 2* presents the relative operator TCO covering all capital expenses and operating expenses for an urban local RAN area in a high-cost market. Different regions and customers have variations in cost structure. Local deviations can be significant, leading to reduced differences but with the same relation in the relative cost structures. The largest cost components are transport rent cost, site rental, energy consumption and radio/RAN equipment. The graph indicates that using self-built transport in the local RAN area is a much more cost-efficient approach than using a leased line to every site, both in DRAN and CRAN architectures. The cost difference is especially large in high-cost markets.

The reason for this is that the introduction of 5G NR significantly increases the radio bandwidth compared with previous generations, which results in increased transport bitrate demands. While typical transport bandwidth to a radio site ranged from 10s to 100s of Mbps in 2G-4G, it is typically up to multiple gigabits per second in 5G. In the lower range of the bandwidth scale, the traditional leased line cost has been manageable. But at sites where the required transport bitrate reaches gigabits-per-second rates, the relative cost for the leased line increases dramatically, accounting for as much as 70-80 percent of the RAN area TCO.

The second largest cost in the "DRAN with leased line to every site" example (and the largest in the other two examples) is site rental. Some scenarios will require densification with new sites, which could be a mix of both macro sites and smaller site types (street sites). However, network densification is likely to face challenges due to the high cost of site rental and limited site availability.

> USING SELF-BUILT TRANSPORT IN THE LOCAL RAN AREA IS A MUCH MORE COST-EFFICIENT APPROACH

There are, however, ongoing discussions in several regions about regulating the high site rental fee for antenna sites, which would significantly increase the opportunity to densify with new sites. The clear trend of tower companies taking over the operation of physical sites and offering site sharing may also decrease site rent cost.

RAN equipment and energy rank as the third and fourth largest costs in all three examples. These cost components are dependent on the deployed RAN architecture. Due to different prices in different markets and areas, DRAN is more cost-efficient in some cases, while CRAN is in others. This explains why the choice may differ between MNOs.

Leased line versus dark fiber

Leased line is a high value type of service and the fee increases with the required bitrate, making it a big challenge for 5G RAN, as the needed transport bitrates are much higher than in previous generations. White fiber has basically the same cost challenges as leased lines, because it is a service with a Service Level Agreement.

Terms and abbreviations

CN – Core Network | **CO** – Central Office | **CPRI** – Common Public Radio Interface | **CRAN** – Centralized RAN | **CU** – Central Unit | **DRAN** – Distributed RAN | **DU** – Distributed Unit | **eCPRI** – Evolved CPRI | **F1** – Interface CU – DU | **gNB** – GNodeB | **HLS** – Higher-Layer Split | **IAB** – Integrated Access and Backhaul | **ISD** – Inter-Site Distance | **LoS** – Line-of-Sight | **MNO** - Mobile Network Operator | **NG** – Interface gNB - CN | **NR** – New Radio | **RBU** – Radio Base Unit | **S1** – Interface eNB - CN | **SFP** – Small Form-factor Pluggable Transceiver | **TCO** – Total Cost of Ownership | **UE** – User Equipment | **VRAN** – Virtualized RAN

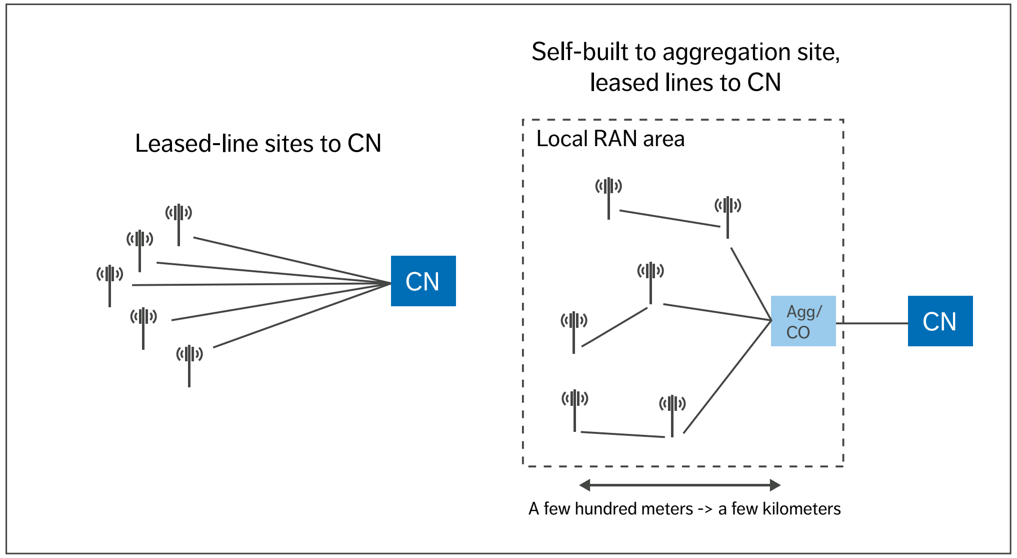

Figure 3 Traditional leased-line approach versus self-built transport in local RAN area

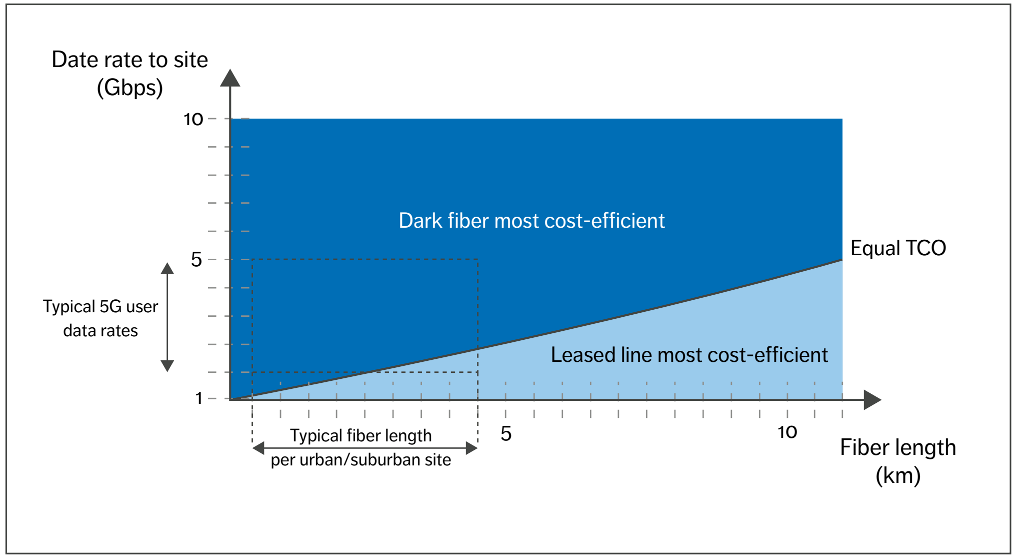

Figure 4 Relative costs for leased lines and dark fiber

Dark fiber rental also has a rather high cost structure, but the transport fee is independent of bitrates and instead based on the fiber distance. Dark fiber solutions therefore fit well in RAN areas with short distances and are preferably deployed, so that the same fiber can be shared, to some extent, by multiple sites. *Figure 3* illustrates the difference between a traditional leased-line approach and self-built transport based on dark fiber. *Figure 4* shows which of these two transport solutions is most cost-efficient depending on data rate to site and site distance.

A self-built transport network based on dark fiber may be deployed with different fiber and radio site structures such as star, subtend or ring topology. The most cost-efficient topology is subtending, where multiple sites share fiber. If network resiliency is required, a ring topology is suitable at the expense of greater fiber length. A pure star topology gives maximum resilience but has the greatest fiber length and is therefore the most expensive choice.

Figure 4 illustrates the typical fiber length per site, where the shortest lengths appear in urban areas using the subtending topology, and the longest distances in suburban areas using the star topology. Figure 4 also shows the typical user data rates for 5G. Dark fiber is more cost-efficient than leased lines in denser areas where the fiber length per site is low, and the data rates are high. If the fiber length becomes longer, or the data rates are smaller, leased lines are more cost-efficient.

For the different technology options on top of dark fiber, the passive optical solution is the most cost-efficient self-built optical solution. This assumes that the number of sites and equipment subtended on the fiber is within the scaling of wavelengths in the system.

The alternative self-built packet-based solution has the advantages of statistical multiplexing throughout the network and can be an L2 Ethernet switched and/or L3 IP routed solution. It assumes that all radio equipment supports a packet-forwarding interface.

Alternatively, when dark fiber is not available or too costly, wireless transport such as IAB or microwave links may be used. These require line-of-sight (LoS) or near-LoS.

Conclusion

Our analysis indicates that due to the large increase in required bit rate per site for 5G NR, the use of traditional leased lines as transport to every radio/antenna site in the RAN will be associated with a high cost in denser areas. Self-built transport in the RAN area is a significantly more cost-efficient alternative for mobile operators. Dark fiber is one self-built transport alternative; microwave links is another.

Since dark fiber cost scales with distance rather than bandwidth, and the trend with 5G is toward shorter site-to-site distances and higher bit rates, dark fiber will be significantly more cost-efficient than leased lines in many scenarios. Further, the large number of fiber providers has boosted availability and competition, resulting in a decrease in fiber rental cost in most urban areas, as well as in some suburban ones. Beyond the RAN area where the local traffic is aggregated and self-built transport is terminated, traditional leased line services to the mobile core continue to be a reasonable solution.

Distributed RAN (DRAN), which works well over both fiber and wireless transport solutions, will continue to be the dominant deployment architecture in most situations. Centralized RAN (CRAN) is an interesting deployment architecture for regions or high-traffic areas where dark fiber transport is available. CRAN offers operational benefits by pooling all baseband to a central site, which results in potential cost savings in site rental and energy, and maximizes the opportunity for inter-site coordination features. In cases where a network has an existing cloud infrastructure, the operator may benefit from adding a high-layer split virtualized RAN deployment to a DRAN or CRAN architecture.

Because the flexibility of the 5G NR architecture enables much greater distribution of equipment and sites than ever before, it is necessary to simplify the installation, deployment and operation of both the RAN and its transport. A high degree of automation and tight integration between the two will be critical to achieving cost-efficient deployments.

> … AUTOMATION AND TIGHT INTEGRATION WILL BE CRITICAL TO ACHIEVING COST-EFFICIENT DEPLOYMENTS

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Further reading

》 **Learn more about building 5G networks at:** *https://www.ericsson.com/en/5g/5g-networks*

5G NR RAN & TRANSPORT OPTIONS ✱

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NEXT-GEN EDGE-CLOUD ECOSYSTEM ✱

CREATING THE

Next-generation edge-cloud ecosystem

Edge computing has great potential to help communication service providers improve content delivery, enable extreme low-latency use cases and meet stringent legal requirements on data security and privacy. To succeed, they need to deliver solutions that can host different kinds of platforms and provide a high level of flexibility for application developers.

PÉTER SUSKOVICS,
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As well-established, trusted partners that already provide device connectivity, mobility support, privacy, security and reliability, the telecommunications industry and communication service providers (CSPs) more broadly have a competitive advantage in edge computing. This advantage is compounded by their ability to reach out globally to all edge sites with relative ease.

The main benefit of edge computing is the ability to move workloads from devices into the cloud, where resources are less expensive and it is easier to benefit from economies of scale. At the same time, it is possible to optimize latency and reliability and achieve significant savings in network communication resources by locating certain application components at the edge, close to the devices. To efficiently meet application and service needs for low latency, reliability and isolation, edge clouds are typically located at the boundary between access networks or on-premises for local deployments.

Since its invention a decade ago, edge computing has mainly been used to improve consumer QoE by reducing network latency and potential congestion points to speed up content delivery. It also lowers operator costs by reducing peering traffic. Now, as a result of the surge in data volume that will come from the massive number of devices enabled by New Radio, the rollout of 5G has made edge computing more important than ever before.

Beyond its abilities to reduce peering traffic and improve user experience in areas such as video, augmented reality, virtual reality, mixed reality and gaming, edge computing also plays a key role in enabling ultra-reliable low-latency communication use cases in industrial manufacturing. It also helps operators meet stringent legal requirements on data security and privacy that are making it increasingly problematic to store data in a global cloud.

Edge-computing applications will have differing requirements depending on which driver has motivated them, and they will be built around different ecosystems that utilize platforms that may be ecosystem-specific. For example, the platforms and application programming interfaces (APIs) for smart manufacturing are different from those required for gaming and other consumer-segment-related use cases, which can be based on web-scale platforms and APIs. A robust edge-computing solution must be able to host platforms of different kinds and provide a high level of flexibility for application developers.

Key factors shaping the edge-cloud ecosystem

On top of being able to meet the requirements of emerging 5G use cases, there are other important factors to consider when designing an edge-computing solution, namely:

- Application design trends, life-cycle management and platform capabilities
- Expectations on management and orchestration
- Edge-computing industry status.

Application design trends, life-cycle management and platform capabilities

Cloud-native design principles have become a common design pattern for modern applications – both for telecom workloads [1] as well as other services. The modular, microservice-based architecture of cloud native applications enables significant efficiency gains and innovation potential when paired with an execution environment and a management system designed to handle cloud-native applications.

Reuse of generic microservice designs across different applications and enhanced platform services allows developers to focus on core aspects of the service with regard to quality and innovation.

Edge computing

Edge computing is a form of cloud computing that pushes the data processing power (compute) out to the edge devices rather than centralizing compute and storage in a single data center. This reduces latency and network contention between the equipment and the user, which increases responsiveness. Efficiency may also improve because only the results of the data processing need to be transported over networks, which consumes far less network bandwidth than traditional cloud computing architectures. The Internet of Things – which uses edge sensors to collect data from geographically dispersed areas – is the most common use case for edge computing.

Hyperscale cloud providers are extending their ecosystem toward the edge, and as part of the Industry 4.0 transformation enterprises are establishing use-case-specific development environments for their edge. The Cloud Native Computing Foundation [2] is gaining traction across all these development ecosystems, enabling portability of applications to private and public clouds.

“ IT IS VITAL TO PLACE ONLY THE APPLICATIONS THAT WILL PROVIDE THE MOST BENEFIT AT THE EDGE ”

The increased amount of individual software modules and the demand to manage them efficiently implies the use of container technology to package and execute those software modules.

Kubernetes has become the platform of choice for container-based, cloud-native applications in both the telecom industry as well as for general-purpose services. Northbound management systems for telecom edge workloads as well as non-telecom edge workloads delegate some life-cycle management functionality to Kubernetes, thus reducing complexity in those management systems.

The Cloud Native Computing Foundation (CNCF) ecosystem has become a focal point for developers aiming to build modern, scalable cloud-native applications and infrastructure. Embracing a certified Kubernetes platform is the best way to become compatible with the CNCF ecosystem and thereby utilize the speed of innovation and variety of applications being developed.

Expectations on management and orchestration

The primary role of management and orchestration is to assure and optimize the application platform, 3GPP-defined connectivity, cloud infrastructure and transport, as well as ensuring the optimal placement of the edge application.

Put in the simplest terms, edge computing is an optimization challenge at scale that consists of several different aspects. The first is supporting the consumer experience by placing appropriate functionality – such as latency-sensitive applications – at the edge. The second aspect is ensuring that the users are connected to these applications. The third aspect is reducing the stress on transport resources and contributing to network efficiency by placing certain types of caching functions at the edge.

While it may seem ideal from a performance perspective to place all applications at the edge, edge resources are limited and prioritizations must be made. From an optimization perspective, it is vital to place only the applications that will provide the most benefit at the edge. Determining the best location for the management functionality – that is, the analytics functionality that can reduce traffic backhaul at the cost of local processing – is a critical aspect of the optimization process. In some cases, local deployment of the management functionality may be necessary to meet service continuity expectations.

A related consideration is the life-cycle management of the edge applications and the edge application platform, which must be efficiently onboarded from a central location, distributed and instantiated to the correct locations. The responsibilities for this can differ depending on the agreement between the edge application platform provider and the CSP. When deploying the edge applications and the edge application platform, appropriate connectivity to both the radio and the broader network must be established.

An edge-computing solution must be able to manage many distributed edge sites that each have their own needs based on local usage patterns. The massive scale that arises from this presents a multidimensional challenge. To overcome it, the

Terms and abbreviations

API – Application Programming Interface | **CNCF** – Cloud Native Computing Foundation | **CSP** – Communication Service Provider | **DNS** – Domain Name System | **IoT** – Internet of Things | **NFV** – Network Functions Virtualization | **ONAP** – Open Networking Automation Platform | **UPF** –User Plane Function | **VNF** – Virtual Network Function | **WAN** –Wide Area Network

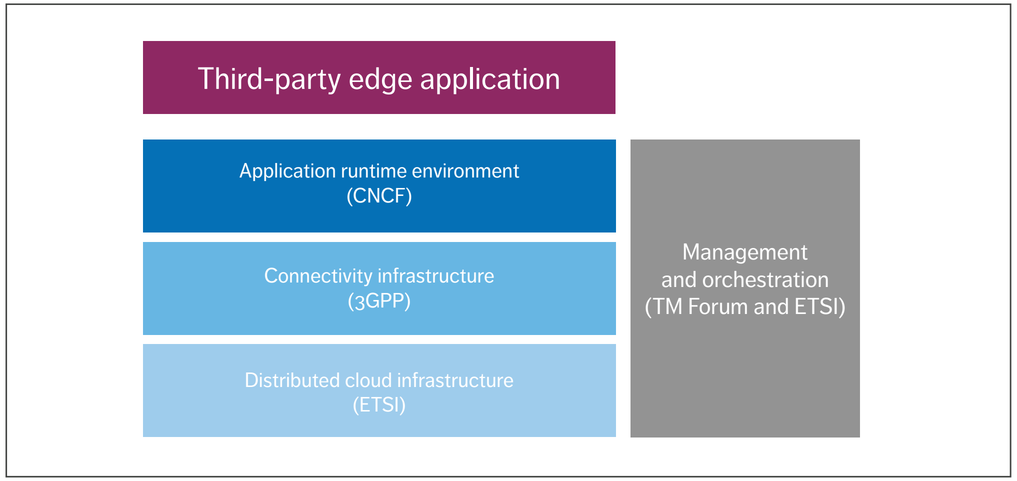

Figure 1 Relevant standardization and open-source forums

different domains of management and orchestration – ranging from hardware to virtualization infrastructure to radio and core network applications, together with edge-application platform orchestration – must all work together in an optimal manner.

Edge-computing industry status

Edge computing is dependent on functionalities in multiple domains. For example, the first step in application deployment is to ensure that runtime is available in the appropriate place, which puts requirements on the orchestration layer and placement capabilities, as well as on business interfaces.

Once the runtime is deployed, anchoring and connectivity are required to configure the necessary local breakout points and steer the traffic to where the edge runtime requires it. Most of these functionalities are not specific to edge computing and have either been addressed by industry standardization or open source. *Figure 1* presents the most relevant standardization and open-source forums for third-party edge applications.

On the networking side, the 3GPP has been addressing edge-computing requirements since release 14, both from the connectivity perspective as well as from a service and exposure perspective. Addressing edge computing under the 3GPP is the only guarantee to secure full compatibility with existing telecommunication network deployments and their future evolution [3].

In the implementation domain, ETSI (the European Telecommunications Standards Institute) Network Functions Virtualization (NFV) [4] defines the infrastructure, orchestration and management, while TM Forum leads the way for the digital transformation of CSPs.

When it comes to runtime and APIs, the fragmentation of the use cases is standing in the way of the vision of one runtime and one type of API. Some developers will use a widely adopted runtime like Kubernetes, especially its versions certified by the CNCF, or embrace web-scale platforms, while some verticals will probably develop, or set requirements on, their own platform and/or APIs. The 5G-ACIA (5G Alliance for Connected Industries and Automation) consortium [5] is one such example. A comparable initiative in the automotive sector is the AECC (Automotive Edge Computing Consortium) [6].

By utilizing standard components and telecommunication infrastructure that is already

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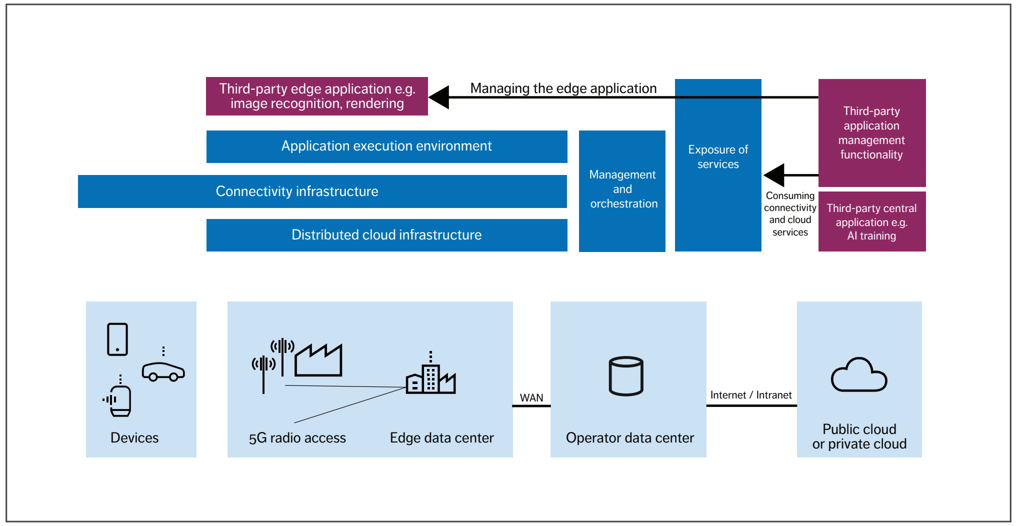

Figure 2 High-level architecture of an edge-computing solution for a typical application

in place, a CSP will be prepared to host any type of third-party application or application platform.

Our high-level solution proposal

Based on our understanding of the key factors shaping the edge-cloud ecosystem, we have defined three main principles that underpin our approach to edge computing:

- Reuse industrialized and proven capabilities whenever possible.
- Ensure backward compatibility.
- Capitalize on existing ecosystems.

The first principle is a reminder that many of the functionalities needed to enable edge computing are not specific to edge computing. They have been used and improved over time, and they should be reused where appropriate. Further, the first principle discourages the adoption of highly specialized solutions early in the process, in light of the current market fragmentation and the uncertainties about the winning use cases in this segment.

The second principle highlights the importance of ensuring that it is possible to deploy existing applications that would benefit from edge deployment without requiring a rewrite on both the device and backend sides.

The third principle pushes us to make the transfer of applications from a central cloud to the edge as transparently as possible to the developers. This means there should be no changes to the life-cycle management of the applications, and existing platforms (along with any specialized ones) should continue to be used for application management and to provide the services the developers need.

With these principles to guide us, we propose a solution with the capabilities to onboard edge applications and edge application platforms into a CSP environment, which can be distributed to the edge data center, central data center or public cloud. *Figure 2* depicts the high-level architecture. The dark-blue boxes represent the main components of our solution and the purple ones indicate third-party applications.

We designed this solution to meet four key criteria:

1. The solution must be able to host different kinds of platforms for different application types.
2. To harmonize with existing developer communities, the execution environment must be CNCF certified (when it is provided by the CSP).
3. To address scaling and mobility issues, the orchestration and management solution of the runtime environment must be aligned with similar functionalities of the network.
4. The solution must both be compatible with 4G and 5G standards and avoid introducing a new layer of complexity (only simple and necessary APIs should be provided).

The solution is based on the distributed cloud infrastructure for virtual network functions (VNFs) and the ETSI NFV orchestration and management functionalities. The same orchestration and management functions are used for the connectivity infrastructure, distributed cloud infrastructure, wide area network (transport) orchestration and the orchestration and management of the application execution environment. This also ensures that there is a user plane function (UPF) available close to the application runtime at the right scaling level that the session management function can select.

To enable transparent connectivity between the edge application and the device, the connectivity infrastructure in our solution is 3GPP compatible. As a result, no edge-solution-specific enhancements are needed in the device.

The exposure functionality provides the main APIs to the third-party developers, of which there are two main types. The first set of APIs is for the business relation with the operator, to enable the onboarding and management of the runtime environment itself and to configure and monitor the connectivity through aggregated APIs built on top of the 3GPP's service capability exposure function, network exposure function and operations support systems/business support systems APIs.

The other set of APIs can be exposed to third-party developers for the deployment and management of the applications themselves. We propose that, for this type of API, a CNCF-certified Kubernetes distribution should be offered in a way similar to how it is provided on web-scale clouds today. This approach harmonizes with the trends and provides developers with greater flexibility.

Runtime environment

To provide a broad baseline for the adoption of applications at the edge, our solution provides customizable Kubernetes distribution in addition to the ability to onboard arbitrary third-party runtime environments.

One of the main benefits of Kubernetes in many different use cases is its modularity. The plugins available in its runtime environment allow a high degree of customization to fit a specific type of workload. We know, however, that industrial applications often rely on dedicated runtime environments that provide tailor-made characteristics, which means that the edge will generally consist of several different runtime environments. As a result, we believe that efficient management of a multitude of different runtime environments is one of the most important capabilities of the edge-computing solution.

Networking and connectivity aspects

Networking requirements in edge deployments are mainly about facilitating connectivity between attached devices and central services (traditional networking), attached devices and edge applications, and edge applications and central services.

The demands on connectivity typically vary between different types of edge applications – both with regard to the type of connectivity as well as the required characteristics. The execution environment, infrastructure, UPFs and management systems must provide the required connectivity services flexibly and efficiently.

Kubernetes provides a variety of container network interfaces to manage connectivity both between microservices and to external endpoints.

Further connectivity features for north-south traffic are enabled by ingress and egress controllers.

The underlying infrastructure is expected to provide basic layer-2 and layer-3 connectivity to support the Kubernetes networking layer. This significantly reduces the management complexity for the underlying infrastructure and bypasses the need to integrate the Kubernetes layer into any lower layer infrastructure management system.

There are several technologies in 4G and 5G to provide local breakout functionality. The packet core VNFs (such as the UPF) can provide local breakout capabilities for the traffic to be routed to the applications in the edge locations. Distributed Anchor Point is a generic solution available today that successfully addresses many use cases and requires no further standardization.

Looking further ahead, Session Breakout is a Domain Name System (DNS)-based solution for dynamic breakout that still needs industry alignment. It is expected to solve issues in many use cases (including enterprise breakout). Session Breakout can provide optimal traffic-routing according to a Service Level Agreement, for example. 3GPP standardization will be needed to address DNS/IP and exposure use.

Multiple Sessions is a target solution that requires further industry alignment and support in devices (iOS and Android, for example). Based on service-peering principles, it would map applications to specific sessions on the user equipment side, thereby meeting the needs of all use cases along with operators' expectations for network slicing.

Network management, orchestration and assurance aspects

Distributed cloud and edge capabilities require the support of several layers in the network: the transport layer, the virtualization infrastructure layer, the access and core connectivity layer and the edge application layer. *Figure 3* shows how these four layers fit together in the context of consumer devices (on the left) and distributed sites (at the bottom). The edge-application platform (the runtime environment) sits between the third and fourth layers, supported by network management, orchestration and assurance.

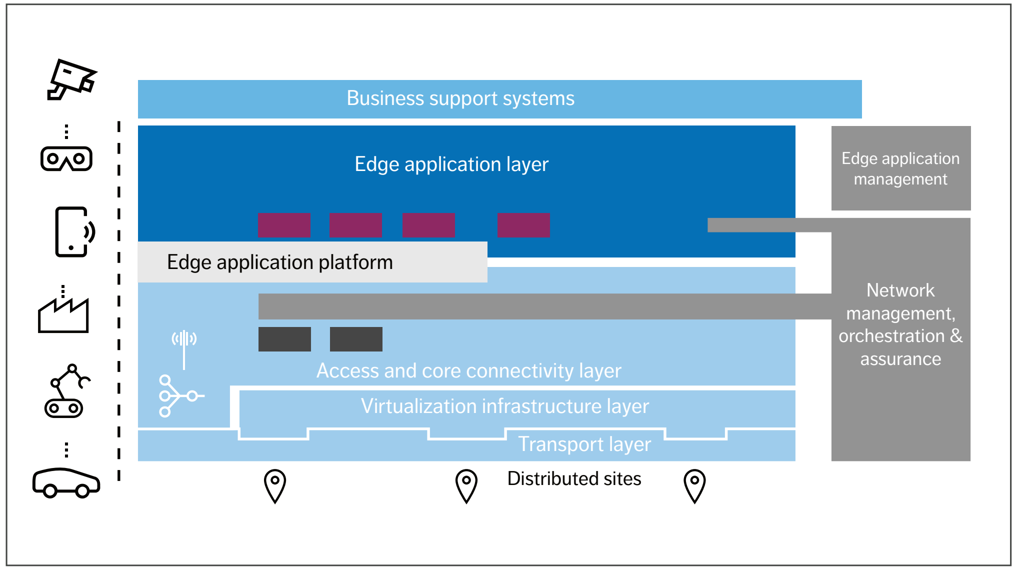

Figure 3 Edge deployment and orchestration

IN THE FUTURE, ORCHESTRATION AND CONFIGURATION CAPABILITIES MAY ALSO BE ABLE TO PERFORM LOCAL HEALING ACTIONS

Several orchestration and management aspects must be considered, particularly with respect to edge applications (the purple boxes in Figure 3), the edge-application platform, VNFs (shown as dark gray boxes in Figure 3), virtualization infrastructure and the distribution of management functionality.

There are two general approaches to handling edge applications. The first is to treat them like an operator's VNF. Third-party edge applications that will be executed on the edge-application platform require a different approach. These applications will be centrally onboarded, then distributed, and life-cycle managed by the edge platform. When instantiating, the CSP's orchestration and management will create the connectivity to the consumer device over the radio network as well as the connectivity to the internet. The application management and overall assurance can be performed by the edge-application provider or by the CSP.

The edge-application platform(s) can be onboarded and managed by the CSP like a VNF, in the same way that any VNF is onboarded and operated on any virtual infrastructure. The CSP will expose capabilities to instantiate the edge application platform instances when and where required.

VNFs (including cloud-native VNFs) are onboarded, designed and life-cycle managed in a similar way to central data center deployments, with two additional considerations: transport between sites and appropriate distribution of VNFs to the edge to optimize user experience. These capabilities already exist in products based on ETSI Management and Orchestration (MANO) [7] and/or the Open Network Automation Platform (ONAP) [8].

Finally, in a distributed environment, it can be useful to distribute certain management functionalities such as analytics and artificial intelligence functions that can perform local analysis and deliver processed insights, rather than backhauling unnecessary data to a central server. In the future, orchestration and configuration capabilities may also be able to perform local healing actions to support either efficient edge operations or edge-service continuity, even when communication to the edge site has been lost.

Conclusion

Edge computing will play a vital role in enabling a wide range of 5G use cases and helping service providers meet stringent legal requirements on data security and privacy. Beyond its abilities to reduce peering traffic and improve user experience in areas such as video, augmented reality, virtual reality, mixed reality and gaming, edge-computing is also needed to enable ultra-reliable low-latency communication use cases in industrial manufacturing and a variety of other sectors. To meet these diverse needs, communication service providers must be able to deliver edge-computing solutions that can host different kinds of platforms and provide a high level of flexibility for application developers.

It is our view that successful development of an edge-computing solution requires a solid understanding of the use cases, associated deployment options and application-developer communities. It is of critical importance that the solution is able to onboard third-party applications and/or application environments, utilizing methods defined by operations support systems standardization bodies such as TM Forum. Rather than building a new application ecosystem and platform, we strongly recommend reusing industrialized and proven capabilities, utilizing the momentum created with CNCF, and ensuring backward compatibility.

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Enhancing RAN performance with AI

Artificial intelligence (AI) has a key role to play in helping operators achieve a high degree of automation, increase network performance and shorten time to market for new features. Our research demonstrates that graph-based frameworks for both network design and network optimization can generate considerable benefits for operators. Even greater benefits can be achieved in the longer term through a comprehensive AI-based RAN redesign.

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Advanced 5G use cases and services in areas such as ultra-reliable low latency communications, massive machine-type communications and enhanced mobile broadband place heavy demands on RANs in terms of performance, latency, reliability and efficiency.

The wide variety of network requirements, paired with a growing number of control parameters of modern RANs, has given rise to an overly complex system for which vendors are finding it increasingly difficult to write maintenance, operation and fast-control software. There is a clear need to both simplify the management and provisioning of the different services and improve the performance of the services offered.

The technical objectives of simplification and performance improvement can be roughly mapped to the business objectives of reducing operating and capital expenses respectively, which translate into reduced cost-per-byte for communication service providers and increased QoS for consumers. Embracing AI techniques for the design of cellular systems has the potential to address many challenges in the context of both simplification and performance improvement [1], making it possible to achieve new objectives that are beyond the reach of classical optimization and rule-based approaches.

> EMBRACING AI TECHNIQUES FOR THE DESIGN OF CELLULAR SYSTEMS HAS THE POTENTIAL TO ADDRESS MANY CHALLENGES

In terms of simplification, AI has already shown the capability to significantly improve functionalities such as anomaly detection, predictive maintenance and the reduction of site interventions through automated site inspections with drones. Performance improvement in the RAN is a greater challenge, as it requires the replacement of classic rule-based network functionalities with their AI-based counterparts. Additional requirements include flexible and programmable data pipelines for data collection and storage; frameworks for the creation (training), execution (inference) and updating of the models; the adoption of graphical processing units for training; and the design of new chipsets for inference.

Three domains for RAN performance improvement

Improving RAN performance involves updating the RAN's control parameters across time, frequency and space to adapt the RAN operation to both static network characteristics, such as the 3D geometry of the surroundings and dynamic network changes in channel, users and traffic distributions. A key prerequisite to successfully apply AI in this context is a deep understanding of the nature and role of different classes of parameters affecting network performance, as well as the complexity of and potential to improve each class.

Artificial intelligence

Artificial intelligence (AI) has experienced an extraordinary renaissance in recent years. The abundance of data and computational capacity that are available today have finally made decades-old techniques like deep learning practically feasible. Substantial investments from both the public and private sectors have fueled the growth of an ecosystem comprised of libraries, platforms, publications and so on that has propelled the field forward and facilitated access to AI techniques for practitioners in various areas.

While the theoretical advances of the AI discipline often occur in domains such as image processing and games, the strengths exhibited by the resulting AI systems – such as the ability to optimize across multiple variables and identify patterns over complex time series – have attracted attention in many industries. In finance, manufacturing and logistics, for example, such capabilities show great potential to improve performance, reduce costs and speed up time to market.

Domain	Parameter type	Network entities	Update frequency
Network design	Deployment parameters	Basebands, cells, RAN configurations, and so on	Monthly/ weekly
Network optimization	Network hyperparameters	Cell clusters/ individual cells	Weekly/ daily/hourly
RAN algorithms	L3 to L1 transmission parameters	Cells and user equipment	Seconds/ milliseconds

Figure 1 Main performance improvement domains

Figure 1 illustrates the main domains for performance improvement that we have identified at Ericsson: network design, network optimization and RAN algorithms. The domains are characterized based on the type of parameters involved, the type and number of network entities and the frequency at which updates typically take place.

Network design domain

The network design domain focuses on improving the parameters that define network deployment – such as the number and location of new cells, the associations of cells to baseband (BB) units, the selection of BB units to form an elastic RAN (E-RAN) configuration, and so on. Network design traditionally relies on planning tools and the domain knowledge of engineers and is performed rather infrequently, such as when new cells are added to an existing network.

Network optimization domain

The network optimization domain focuses on tuning network hyperparameters. While the term hyperparameter has been strongly associated with machine learning in recent years, it generally refers to any parameter used to control the behavior of an underlying algorithm. The hyperparameters of the algorithm are tuned to produce, for the same measured input, a different output that is more appropriate for the given scenario.

While network hyperparameters encompass both the core network and the RAN, our focus here is on RAN hyperparameters such as static/semi-static configuration parameters for cells and user equipment as well as the hyperparameters of RAN algorithms.

Network hyperparameters are optimized to slowly adapt the RAN algorithms to different network scenarios and conditions and bring the performance of a certain area of the network (a particular cluster of cells, for example) into a steady state wherein specific key performance indicators (KPIs) are improved. Examples include hyperparameters for self-organizing networks algorithms and L3 algorithms (mobility, load balancing and so on) for coordination algorithms (such as coordinated multi-point (CoMP), multi-connectivity, carrier aggregation (CA) and supplementary uplink), as well as for L1/L2 algorithms (uplink power control, link adaptation, scheduling and the like).

RAN algorithms domain

The RAN algorithms domain focuses on optimizing the L3 to L1 control parameters that directly affect the signal transmitted to/from the user. Examples include handover and connectivity decisions and the allocation to users of resources such as modulation and coding scheme, resource blocks, power and beams. The L3 to L1 algorithms adapt these parameters on a fast timescale, for individual network entities (cells and UEs, for example), to the rapidly changing environment conditions in terms of channel, traffic, user distribution and so on.

Our path toward AI-based RAN optimization

A natural first step toward a wide integration of AI in RAN products for performance enhancement is the adoption of AI-based solutions in the network design and optimization domains. Optimizing the RAN by tuning the network hyperparameters is safer and easier than redesigning the RAN algorithms with AI-based solutions, as it consists of an outer control loop that does not modify the RAN algorithm design itself but only tunes its behavior.

Figure 2 demonstrates how different network hyperparameter values result in different behaviors for the underlying RAN algorithm, which are represented by different shapes. However, the performance improvement achievable by AI-based network optimization remains limited by the underlying design of the RAN algorithms and the frequency at which network hyperparameters c an be adapted, which affects the extent to which the system can be controlled.

At Ericsson, our long-term goal is to create an all-encompassing AI-based framework that spans the full hierarchy of control – that is, not only network design and optimization but also, importantly, AI-based RAN algorithms.

Examples of AI applications in today's networks

Based on our long-standing research in the area of how AI can be used to improve RAN performance, Ericsson has developed powerful AI-based frameworks for network design and network optimization, as well as several other AI-based solutions for specific use cases.

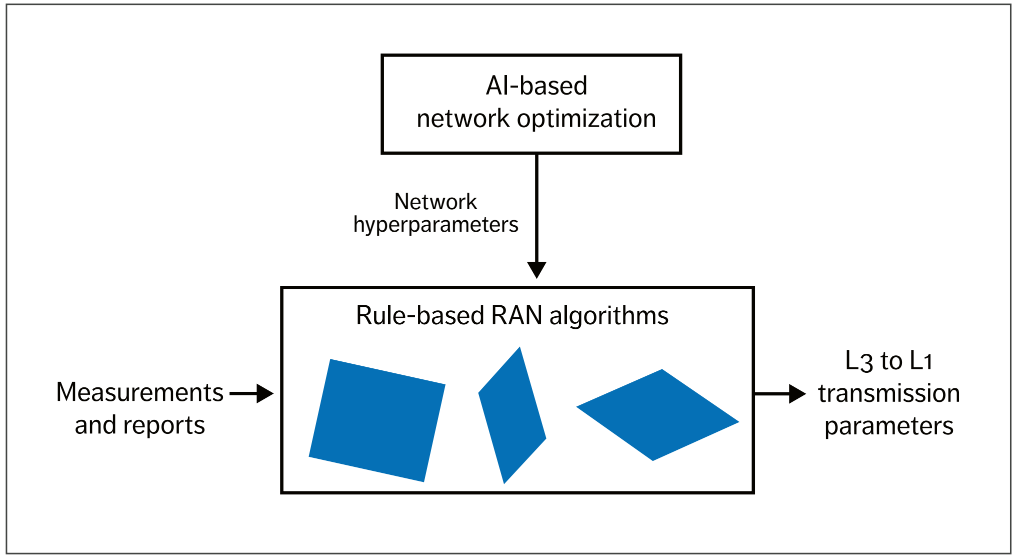

Figure 2 Impact of different hyperparameter values on the behavior of the underlying algorithms

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Network design framework

In both 4G and 5G, our centralized RAN (C-RAN) and E-RAN interconnect BB units to allow optimal coordination across the entire network in a centralized, distributed or hybrid network architecture. To ensure that C-RAN and E-RAN performance is in line with customer expectations, a thorough network (re)design is required. In this regard, AI techniques based on advanced network graph methodologies are applied to understand and characterize the complex radio network and its underlying structures, such as the relations between cells and BB units. This approach leads to an optimal design that maximizes consumer throughput through optimized CoMP and CA techniques, and the design is also future-proof in terms of capacity and technology expansions. The design can be split into two main steps.

In the first step, with C-RAN, BB operation is shifted from site location to a centralized BB hub. The C-RAN design therefore focuses on the reconfiguration of the existing distributed RAN architecture to a centralized architecture, where cells are grouped in a centralized hub. This is done in such a way as to create the optimal coordination among cells belonging to the same BB unit, resulting in higher spectrum efficiency and improved consumer experience.

C-RAN configuration design is a highly complex task and difficult to solve using a traditional network design approach. This is because finding an optimal cell grouping that maximizes network performance among a large number of possible cell grouping combinations requires numerous aspects to be considered, such as:

- Intra and inter-frequency cell coverage overlap and neighbor signal strength
- Signal quality and diversity to improve coordination techniques
- Distance between cells
- Frequency band distribution per BB unit
- BB capacity design
- Future cells/sites deployment.

Using an AI-based network graph analysis, natural and hidden structures within cell relations (also known as communities) can be discovered. Based on the various network indicators listed above, the strength of each cell relationship can be measured by a weight factor. The higher the weight factor, the more likely it is that these cells should be grouped together into the same BB unit.

In the second step, E-RAN enables flexible coordination between BB units irrespective of the BB deployment. Similar to the C-RAN design, an AI-based network graph approach can also be applied here to obtain optimal BB clusters consisting of a set of interconnected BB units for borderless coordination across the entire network.

Terms and abbreviations

AI – Artificial Intelligence | **BB** – Baseband | **C-RAN** – Centralized RAN | **CA** – Carrier Aggregation | **CC** – Component Carrier | **CoMP** – Coordinated Multi-Point | **E-RAN** – Elastic RAN | **KPI** –Key Performance Indicator | **L1** – Layer 1 | **L2** – Layer 2 | **L3** – Layer 3 | **RL** – Reinforcement Learning

E-RAN DESIGN IS ENTIRELY AUTOMATED

Figure 3 shows the performance improvement in a 4G network operated by an Asian operator for three KPIs after an automated E-RAN redesign. The first bar graph indicates that the connections in CA mode using three component carriers (CCs) increased by 30 percent. The middle bar graph shows that the data volume carried by any secondary cell increased by 22 percent, while the bar graph on the right shows that downlink cell throughput increased by 4.3 percent. However, the most valuable benefit is that the E-RAN design is entirely automated and performed in minutes rather than t he months of work that would be required by human experts.

Network optimization framework

The monitoring and control of network performance is traditionally handled by a team of engineers supported by expert systems targeted at optimizing particular areas of the network (typically a cluster of cells). As such, network performance is often optimized by using a mix of manual and automated rule-based instructions combined with predetermined thresholds for each network performance metric. These rules and thresholds are completely based on human observations and expertise.

However, our solutions demonstrate that it is possible to create a fully scalable and automated closed-loop AI-based solution for network optimization consisting of automated network data processing, network issue identification and classification, detailed root-cause reasoning and automated parameter configuration

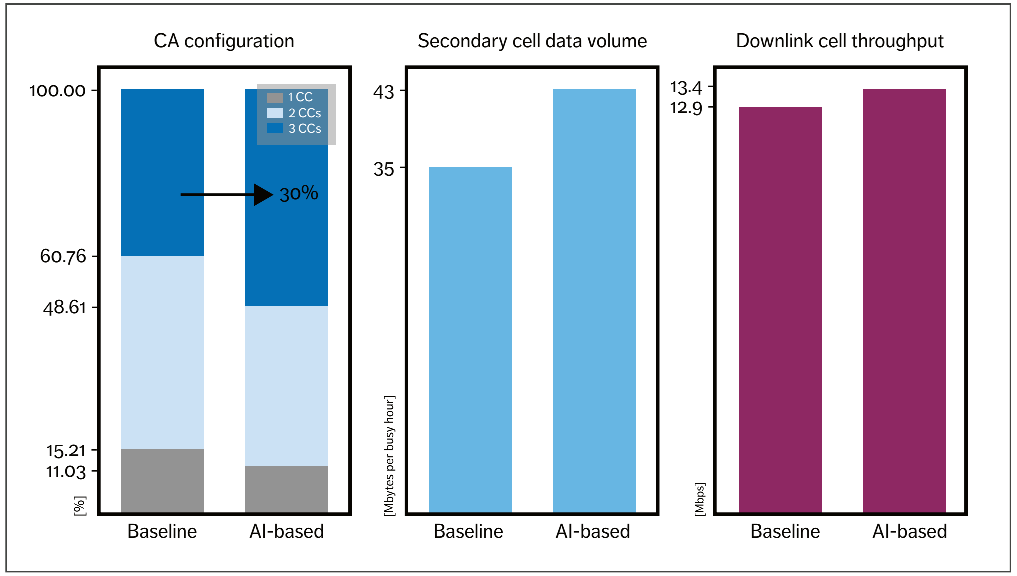

Figure 3 Performance improvement of three KPIs after an automated E-RAN design

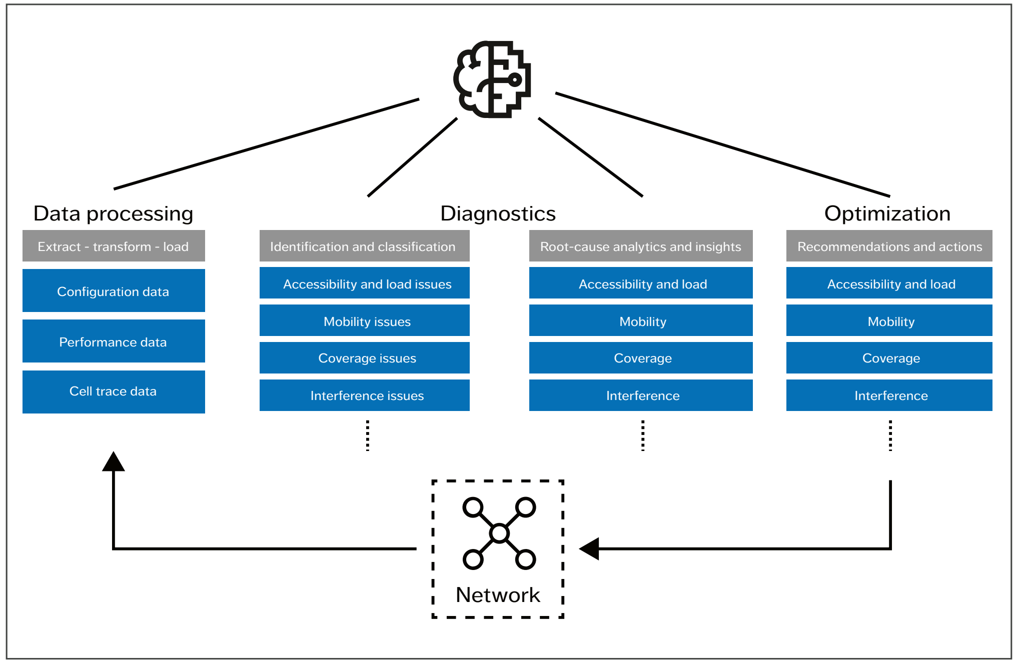

Figure 4 Flow of operations for Ericsson's network optimization framework

recommendations. *Figure 4* illustrates the operations flow for Ericsson's network optimization framework.

State-of-the-art unsupervised and semi-supervised learning techniques combined with expert domain knowledge lead to an efficient annotation of normal and abnormal performance patterns that can be utilized later for issue identification and classification using supervised learning techniques. By integrating network topologies and configurations with hundreds of performance metrics and their two-dimensional correlation in time and space, it is possible to generate a knowledge graph that reveals the specific root causes that lead to an identified network issue.

Closing the automated loop, network parameter changes are automatically suggested to resolve the specific root cause and further improve performance.

AI-based use cases

A non-exhaustive list of AI-based use cases that Ericsson has investigated includes handover [2], link adaptation [3], transmission optimization in C-RAN, interference management, rogue drone detection [4] and federated learning in RAN for privacy awareness [5]. Two of the use cases that are of particular interest in the context of RAN optimization are the prediction of performance on a secondary carrier using primary carrier data [6] and antenna tilting.

ERICSSON CONTINUES TO INVEST SIGNIFICANT R&D RESOURCES IN THE USE OF AI

Secondary carrier prediction

The use of both high-frequency bands such as 28GHz and higher millimeter-wave bands will continue to increase in 5G radio networks and in future generations. A larger number of bands provides higher capacity but results in larger measurement overhead. For instance, initial deployments on the 28GHz frequency bands will provide spotty coverage. For users to be able to make use of potentially spotty coverage on higher frequencies, the UEs need to perform inter-frequency measurements, which could lead to high measurement overhead. We have used AI techniques to predict coverage on the 28GHz band based on measurements at the serving carrier (for example at 3.5GHz). This approach decreased the measurements on a secondary carrier, thus reducing the energy consumption and the delay for activating features like CA, inter-frequency handover and load balancing.

Antenna tilting

AI-based antenna tilting deserves particular attention among network optimization use cases, as it promises to enhance the coverage and capacity of mobile networks by adjusting base station antennas' electrical tilt based on the dynamics of the network environment. Unlike the conventional antenna tilt approach that follows a rule-based policy, AI techniques enable a self-evolving policy, learning from feedback through network KPIs. Using reinforcement learning (RL), an agent is trained to dynamically control the electrical tilt of multiple base stations jointly so as to improve the signal quality of a cell and reduce the interference on neighboring cells in response to changes in the environment, such as traffic and mobility patterns. This results in an overall improvement of network performance and QoE for the users while reducing operational costs.

Next steps

Ericsson continues to invest significant R&D resources in the use of AI in all three RAN performance improvement domains. We expect to see notable advancements in the network design and network optimization domains in the near term, while at the same time we are increasingly shifting our focus to the critically important RAN algorithms domain.

Network design

In the network design domain, we are currently working to make aspects such as cell-to-BB and BB-to-BB connections software defined. This development would enable the integration of automated AI-based network design in a closed loop, where the network continuously reshapes its graph depending on changing traffic patterns or the addition of new nodes to the network.

Network optimization

In the network optimization domain, our near-term goal is to extend the framework to optimize a larger number of hyperparameters at a higher update frequency. In the mid-term, we aim to integrate these new capabilities into our products and ultimately make them a native part of our product offering.

RAN algorithms

Addressing the optimization of the RAN algorithms domain is vital to our long-term vision of creating an all-encompassing single AI-based controller that spans the full hierarchy of control. The benefit of such a controller would be the inherent capability to optimize multiple transmission parameters across layers simultaneously. The creation of a controller

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with the ability to learn directly through exploration of the state space would remove the boundaries imposed by human-designed algorithms, making it possible to identify better combinations of transmission parameters within a layer and across layers. Moreover, a controller with the ability to learn from data would inherently be tuned to the environment and be free of network hyperparameters, which would lead to simplification of the software stack.

Nonetheless, replacing L3 to L1 RAN algorithms with a single AI-based controller presents more challenges than network design and optimization, on several levels. One challenge is that fast parameter changes introduce the problem of transients, and therefore require the AI controller to predict the short-term state evolution of the system due to channel and traffic changes, for example, as well as the actions the AI controller itself submits to the system.

Another challenge is the need to redefine the radio access problem in a way that enables learning through interaction with the RAN environment. Today's divide-and-conquer approach for providing radio access to UEs by breaking down the problem into many subproblems of manageable complexity, and designing specific solutions for each subproblem, is difficult to apply when using AI-based controllers. In other words, staying within the current fragmented RAN framework with different AI-based controllers, each trying to optimize a RAN feature while learning through interaction with the same RAN environment, would prevent the system from learning and jeopardize system performance.

One possible approach to address such challenges would be to adopt RL as the framework of choice for RAN control. RL has the necessary capabilities to deal with transients, but it remains challenging to deploy it in the context of the current fragmented RAN framework. To this end, one approach would be to redefine the problem and devise a solution with a single stage of state estimation and a single stage of downstream end-to-end control. This design choice would enable a state estimation as close as possible to the true system state and a controller capable of joint optimization over several transmission parameters.

Additionally, a true AI-based redesign of the system would require end-to-end integration of the different layers of the control hierarchy in such a way that slower (higher) layers of control (such as network design) can make decisions to improve overall system performance as a function of the models learned for the faster (lower) layers of control (such as RAN algorithms).

Conclusion

Interest in artificial intelligence (AI) is growing rapidly in the telecom industry as operators look for ways to automate RAN operations, boost network performance and shorten the time to market for new features. It is important to note, however, that the successful use of AI to optimize the performance of a radio communication network requires a deep understanding both of the nature and role of the different classes of parameters that affect network performance, as well as the complexity and optimization potential of each class.

At Ericsson, our long-term aim is to redefine the overall concept of radio access control with the intent to create a cellular network that constantly adapts itself to the static and dynamic characteristics of the scenarios as well as the requirements of the customers. To help get us there, we have identified three main RAN performance improvement domains based on the type of parameters involved, the type and number of network entities and the frequency at which updates typically take place.

Our work demonstrates that graph-based frameworks for both network design and network optimization can generate considerable benefits in terms of improved performance, simplified management and shorter time to market. Looking further ahead, we expect that the creation of a single AI controller that replaces RAN algorithms will play a key role in a comprehensive AI-based RAN redesign and ultimately make it possible to achieve performance targets that are unreachable in a traditional rule-based design.

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5G migration strategy

FROM EPS TO 5G SYSTEM

For most mobile operators, the introduction of the 5G System (5GS) will be a migration from their existing Evolved Packet System (EPS) deployment to a combined 4G-5G network that provides seamless voice and data services. This migration requires a carefully tailored, holistic strategy that includes all network domains and considers the operator's specific needs per domain.

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Introducing the 5GS to provide mobile broadband (MBB) in an existing 4G EPS network has significant impacts across all network domains – from the RAN, to packet core, user data and policies, and services – as well as backend systems.

■ The EPS is primarily used today for a variety of MBB use cases. In some cases, EPS deployments have already been upgraded for early support of 5G by non-standalone (NSA) New Radio (NR). Many such 4G and NSA NR operators have already decided to introduce – or are considering introducing – the 5GS as standardized by the 3GPP.

The 5GS introduces support for NR standalone (SA) [1] and is specified to support existing MBB use cases as well as new and improved ones. Ericsson believes that operators can address increasing traffic demands and quickly introduce innovative new services by using NSA NR and SA NR [2].

During the migration period when NR coverage is being built out, services requiring wide-area coverage are best supported through interworking between the 5G Core (5GC) and the existing Evolved Packet Core (EPC). Over time, a growing number of new use cases will utilize the established 5GS MBB scale and wide-area network build-out.

The interworking with the EPC places dependencies on the backend business support systems (BSS) system integration, since user data and policies need to support two networks (the EPC and 5GC). The new devices must support 5GS-related capabilities, while at the same time devices that only support the EPS – including inbound roaming devices – will exist for a long period and will require corresponding network support. This long-term need is a strong argument in support of the concept of a dual-mode core network solution that includes both EPC and 5GC functionality.

A key benefit of a dual-mode core network solution is the common operational model for the EPC and 5GC, which simplifies the management of the overall system. It also includes more granular life-cycle management of the individual software modules (also called microservices) based on cloud-native deployment and operational principles [3]. The common operational model can be used for dynamic and flexible scaling of individual micro-services based on capacity needs, such as rebalancing the EPC versus 5GC resource usage when the device fleet evolves over time from 4G to 5G.

5GS architecture for interworking with EPS

Introducing the 5GS to a network requires a comprehensive strategy that considers all network domains, coverage strategy, spectrum assets and devices, as well as which services to offer where. Among other things, the 5GC, the packet core in the 5GS, introduces new network functions and interfaces internally and toward operations support systems and BSS, including charging systems. It also has new interfaces and protocols toward the next-generation RAN (NG-RAN) and devices, which means that the RAN migration, including spectrum assets and device strategies, needs to be coordinated with the 5GC introduction.

Additionally, any plan to introduce the 5GC must consider its new service-based architecture, which includes a network repository function for service registration and discovery, as well as new capabilities like network slicing support and network exposure.

Operators with both NR and LTE access are able to use 5GC capabilities for tight interworking to the EPS, also known as EPC-5GC tight interworking in the first release of 5G specifications in the 3GPP [4].

Terms and abbreviations

5GC – 5G Core | **5GS** – 5G System | **AMF** – Access and Mobility Management Function | **BSS** – Business Support Systems | **CA** – Carrier Aggregation | **CAS** – Customer Administration System | **CP** – Control Plane | **CUPS** – Control Plane User Plane Separation | **EPC** – Evolved Packet Core | **EPS** – Evolved Packet System | **E-UTRAN** – Evolved Universal Terrestrial Radio Access Network | **FDD** – Frequency Division Duplex | **gNB** – NR Node B | **HSS** – Home Subscriber Server | **HTTP** – Hypertext Transfer Protocol | **IMS** – IP Multimedia Subsystem | **MBB** – Mobile Broadband | **MME** – Mobility Management Entity | **NAS** – Non-Access Stratum | **NG-RAN** – Next-Generation RAN | **NR** – New Radio | **NSA** – Non-Standalone | **PCF** – Policy Control Function | **PCRF** – Policy Control and Charging Rules Function | **PDN** – Packet Data Network | **PDU** – Protocol Data Unit | **PGW** – Packet Data Network Gateway | **REST** – Representational State Transfer | **SA** – Standalone | **SGW** – Serving Gateway | **SMF** – Session Management Function | **SMSoIP** – SMS-over-IP | **SMSoNAS** – SMS-over-NAS | **SPR** – Subscription Profile Repository | **TDD** – Time Division Duplex | **UDM** – Unified Data Management | **UDR** – Unified Data Repository / User Data Repository | **UE** – User Equipment | **UP** – User Plane | **UPF** – User Plane Function

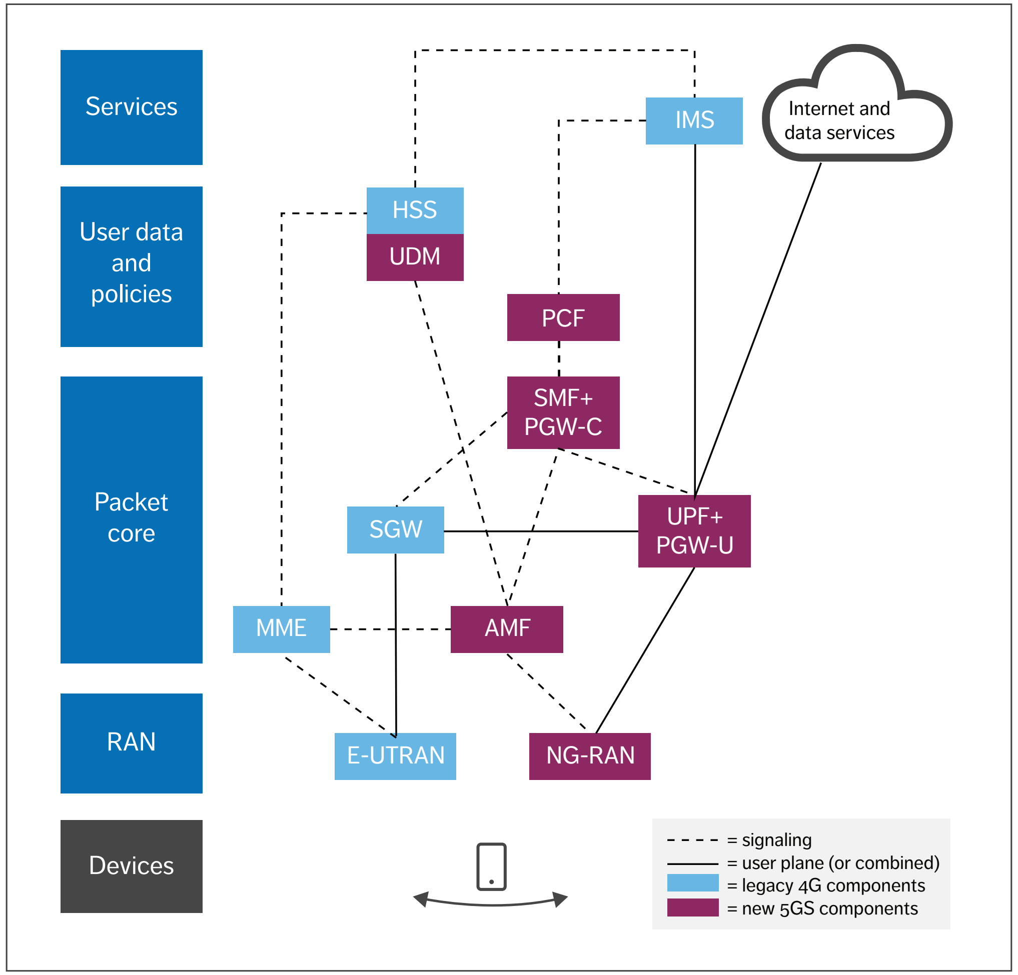

Figure 1 EPC-5GC tight interworking architecture

Figure 1 shows the 5GS architecture for EPC-5GC tight interworking. To enable IP address preservation when connecting over and changing between 4G and 5G access, the 5GC architecture includes a common user plane (UP) anchor point realized by the session management function plus packet data network gateway control plane function (SMF+PGW-C) and the user plane function plus PGW user plane function (UPF+PGW-U).

To support seamless service continuity and network-controlled handover, the Mobility Management Entity (MME) and the new access and mobility management function (AMF) interact directly through the N26 reference point, which supports devices in single-registration mode (the device is either registered in the MME or in the AMF, but not in both simultaneously). The N26 reference point is used for both idle mode and connected mode mobility; the device initiates idle mode mobility (possibly triggered by the RAN), while connected mode is initiated by the RAN, and the device is informed when handover preparation has been completed. Furthermore, tight interworking also includes how to map protocol data unit (PDU) sessions in the 5GS to packet data network (PDN) connections in the EPS and vice versa.

The interworking architecture ensures that the new 5GC-capable devices are always connected to the UPF in the 5GC independently if they are connected though 4G or 5G access, which enables IP address preservation when devices move between accesses. Thus, service characteristics are maintained, since any colocated value-added services connected to the UPF are the same, and the policy control function (PCF) applies session policies for the device when connecting over 4G or 5G access.

Policy and subscription management need to be provided in a consistent way for a device using NR or LTE access. The interworking architecture provides several 5GC capabilities also over LTE/EPC, including support for network slicing. Operators can migrate from an existing EPC to a dual-mode EPC and 5GC network solution by migrating the packet gateway, the policy control and the subscription and data management, and by introducing new functionality.

> POLICY AND SUBSCRIPTION MANAGEMENT NEED TO BE PROVIDED IN A CONSISTENT WAY

Overcoming domain-specific migration challenges

A solid EPS to 5GS migration strategy will consider and address the challenges in devices and all four network domains: RAN, packet core, user data and policies, and services.

The RAN domain

In many markets, the new NR spectrum is first available in mid and high bands. Depending on the site grid, introducing the NG-RAN with NR SA only on these bands may lead to spotty NR coverage that is only suitable for local area services. When deploying NR SA for MBB, it is preferable to ensure continuous NR coverage within the targeted service area (a city, for example) to avoid frequent mobility between NR and LTE. Alternatively, intersystem mobility could be limited by conservative mobility thresholds.

Achieving higher capacity and continuous coverage requires a combination of NR on mid and high bands for capacity and NR FDD on sufficiently low bands for coverage [2]. The NR FDD spectrum on low band can be either new, re-farmed or an existing LTE band that is shared between NR and LTE using dynamic spectrum sharing. NR bands can be combined using carrier aggregation (CA) or, in some cases, dual connectivity.

NR CA will be vital in enabling service providers to serve the growing number of 5G devices in the network while maintaining overall network performance and user experience. This is done by activating downlink CA (FDD+TDD) in the areas with low-band and mid-band NR. This not only boosts mid-band NR coverage, and consequently capacity gain, but also provides a further coverage boost by enabling some of the NR signaling to be moved to lower bands. Ericsson has shown that this can provide up to 3-7dB extra gain in link budget on the downlink [8].

NR SA with CA reduces complexity in the RAN and devices compared with dual connectivity as in NR NSA. The device does not need to transmit on two uplinks at the same time. The use of NR SA also reduces the time from a device being in inactive mode to full NR capacity, enabled by all control signaling being carried over NR instead of being dependent on LTE and the setup of dual connectivity. The benefit to the consumer is faster access to the full potential of the combined NR capacity when, for example, downloading a file or starting up a video.

To support existing and forthcoming services like voice and emergency when migrating MBB to the 5GS, the NR SA needs to support capabilities and coverage demands like intersystem handover, positioning and QoS. This can be a stepwise migration.

The user data and policies domain

The EPC-5GC tight interworking architecture shown in Figure 1 assumes common subscription management support regardless of the access technology used by a given user. Although a combined HSS/UDM (Home Subscriber Server/ Unified Data Management) function is depicted, first deployments support subscription management for the EPC and 5GC by interworking between a separate HSS and UDM through an HTTP/REST (Hypertext Transfer Protocol/Representational State Transfer) interface.

Existing HSS functionality must be evolved to enable interworking with UDM and to support tight interworking between the EPC and 5GC. The evolution includes an upgrade of the HSS functionality offered to EPC serving nodes with, among other things, subscription parameters that enable user access to the 5GC to ensure IP session continuity and single registration across the EPC and 5GC.

Once the 5GC for MBB is introduced, sessions are anchored in the SMF+PGW-C function. The use of SMF+PGW-C allows the policy control and charging rules function (PCRF) used for policy control in the EPC to be replaced by a new dual-mode policy management system that supports 5G-enabled devices regardless of the access technology currently used.

HSS/UDM and PCF/PCRF business logic are also highly dependent on how subscription data and policy subscription data is managed (that is, provisioned, stored and accessed). The EPC allows the support of a data-layered architecture, where the HSS and PCRF make use of an external database to manage subscription data. These databases are known as the User Data Repository (4G-UDR) for HSS subscription data and the Subscription Profile Repository (SPR) for PCRF policy subscription data. The 5GC generalizes the concept into a Unified Data Repository (5G-UDR). The 5G-UDR stores subscription, policy subscription, application and exposure data.

During the introduction of the 5GC it is beneficial to deploy new dual-mode subscription, data and policy management systems that support the EPC-5GC tight interworking architecture and procedures as depicted in *Figure 2*.

The dual-mode data management system enables both subscription data centralization and single point of provisioning in new 5G-UDR instances

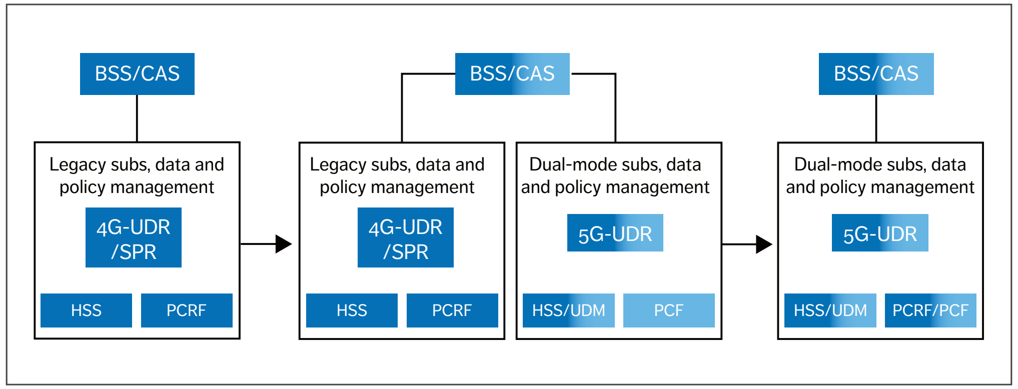

Figure 2 Migration to dual-mode subscription, data and policy management systems

> THE DESIGN OF THIS SOLUTION FOLLOWS THE PRINCIPLES OF A COMMON OPERATIONAL MODEL BASED ON CLOUD-NATIVE IMPLEMENTATION

for 5G-enabled subscribers. This requires subscription data migration from the legacy 4G-UDR/SPR to the 5G-UDR. This migration process can be assisted by automatic migration tools and/or by auto-provisioning/activation mechanisms enabled by notifications to the BSS/CAS (customer administration system), such as the detection of a 4G-only user using a 5G-capable device and/or being active in the 5GC.

As a first migration step, the HSS functionality in the legacy subscription management system may still be used for 5G-enabled users when they connect through the EPC. The existing HSS instances may reach subscription data for 5G-enabled users from the 5G-UDR through the local 4G-UDR/SPR using proprietary interfaces, for example. The PCF in the dual-mode policy management system supports 5G-enabled users even if they connect through the EPC.

Alternatively, 5G-enabled users may be served by HSS functionality in the dual-mode subscription management system when they connect through the EPC. The dual-mode subscription, data and policy management systems for 5G-enabled subscriptions can coexist with the existing subscription, data and policy management systems for legacy 4G subscriptions (existing HSS and PCRF instances and 4G-UDR/SPR) with the support of a signaling routing function.

In either case, the goal is that all types of subscriptions (even legacy 4G-only subscriptions) are managed on the new dual-mode system for subscription, data and policy management, as shown to the right in Figure 2. The design of this solution follows the principles of a common operational model based on cloud-native implementation and offers its benefits to manage both new 5G-enabled subscribers regardless of the access technology used and legacy 4G users who connect only through the EPC.

The packet core domain

According to our definition, the packet core domain includes functionality to handle access and mobility management (MME, AMF), session management, the serving-gateway-control plane function (SGW-C), PGW-C, SMF and UP functionality (SGW-U/PGW-U, UPF).

The initial introduction of the 5GC for wide-area services allows a migration of RAN and core independently of each other, similar to the transition from 3G to 4G. Prior to the standardization release of 5G, the 3GPP also standardized a separation of the gateway functions in the EPC into control plane (CP) and UP, known as CUPS. This separation enables new opportunities for UP distribution and edge breakout of traffic already in the EPC. Separate PDN connections can use different SGW-U/PGW-Us in central and local deployments, for example. The functional separation of the CP and UP in the EPC CUPS is carried over in the 5GC architecture corresponding to the SMF and UPF.

The availability of both CUPS and EPC-5GC tight interworking enables multiple possible migration paths from the EPC to the 5GC. One option is to first introduce CUPS into the EPS, which enables the operator to use the CP and UP separation before migrating to the 5GS. This can be beneficial to handle increased traffic demand when introducing NR NSA and prepare for a smooth migration to the 5GC based on a UP implementation that supports both the EPC and 5GC.

Another option is to introduce CUPS at the same time as the 5GC by colocating SMF+PGW-C with SGW-C functionality and UPF+PGW-U with SGW-U functionality. This allows the new high-capacity 5G devices to be served by a CP and UP split-gateway architecture when connecting over either 4G or 5G access, as shown in the middle

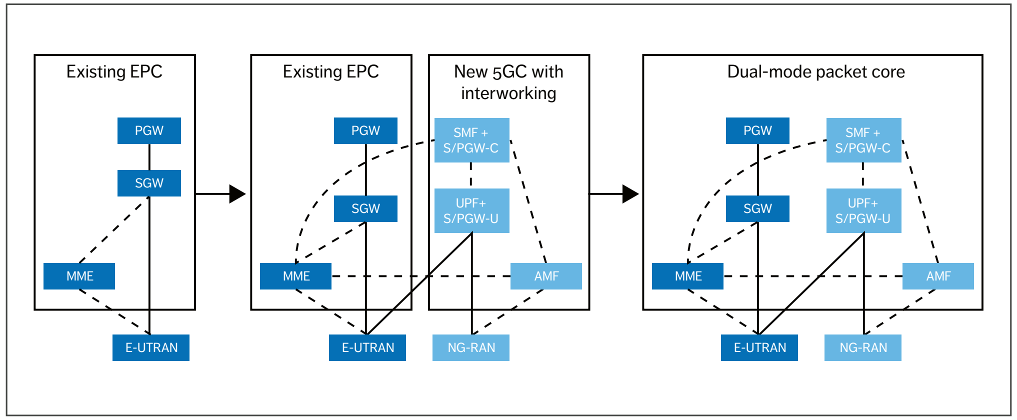

Figure 3 Migration from the EPS to the 5GS utilizing EPC-5GC tight interworking

part of *Figure 3*. An additional benefit of this colocation is the possibility for more flexible distribution of the UP in different locations, for example to address low-latency services by placing the UP closer to the RAN. The 5GC tight interworking can also interwork with a legacy SGW as shown in Figure 1.

The middle part of Figure 3 also shows that it is possible to continue to use the existing SGW and PGW functionality in the EPC for legacy devices with 4G-only subscriptions when introducing the 5GC, which minimizes the impact on existing services and subscribers. However, the MME needs to support the gateway selection (SGW-C & SMF+PGW-C) for devices with 5GC subscription and 5GC-NAS (non-access stratum) capability. The latter is communicated as part of the tracking area update/attach procedure in the EPS. The MME can also utilize additional methods to assist in the gateway selection, such as dedicated Access Point Names or Domain Name System lookup enhancements. The MME may also need to support restrictions, such as if the operator wants to prevent mobility to the 5GS for certain subscribers.

The next migration step is to combine the full EPC and 5GC functionality into a dual-mode packet core, serving 5G-enabled subscribers regardless of the access technology used and legacy 4G users that connect only through the EPC, as shown on the right side of Figure 3. In this example, the new 5GC devices are served by the SMF+PGW-C, while the legacy 4G-only devices are still served by a separate PGW instance. Other deployment models are also conceivable.

The goal of the migration is therefore a solution that follows the principles of a common operational model based on cloud-native implementation and offers its benefits to both new 5G-enabled subscribers regardless of the access technology used and to legacy 4G users, who connect only through the EPC.

When introducing the 5GS, it is also important to consider future services beyond MBB. One way to future-proof the network architecture is through support for network slicing. In this context, all functions required to support MBB and needed for access and mobility management, session management and the UP (SMF+PGW-C and UPF+PGW-U) are part of an MBB network slice. Hence, a single network slice can be used both for internet access and for IMS voice and other IMS services. This arrangement can be maintained when the device starts on the EPC and moves to the 5GC or vice versa.

The services domain

Regardless of access technology, support for IMS voice, emergency services and SMS is expected to work seamlessly anywhere subscribers can connect to the network. A phone will not select NR SA access unless it detects IMS voice service. Voice and emergency services are supported by specific capabilities in the RAN and Packet Core that must be provided to the combined EPS and 5GS after migration. The IMS needs to be updated to support NR SA.

EPS fallback can be used as a first voice migration step when introducing the 5GS in deployments where NR coverage has underlying LTE/EPC coverage. EPS fallback means that the device is moved to EPS at call establishment. It is typically used prior to the deployment of all the voice capabilities in NR or before the RAN is dimensioned and tuned for voice.

The subsequent voice migration from EPS fallback to voice over NR is achieved by allowing voice calls to be established on NR connected to the 5GC instead of performing EPS fallback at call establishment. This migration step can be done once all required voice-over-NR capabilities are in place in the device and the network, and RAN is dimensioned and tuned for voice. However, devices introduced before this step will remain in the field when voice over NR is introduced, which is why the network must support voice over NR including EPS fallback.

With respect to emergency services, the 3GPP has specified different methods for emergency calls in the 5GS. For example, using a service request to perform EPS fallback of emergency calls is a possible first migration step. The benefit of this approach is that it only impacts the AMF and gNB, and it avoids regulatory requirements on the 5GS related to emergency calls. Migrating to emergency-call-over-NR requires all regulatory requirements related to emergency calls to be addressed. Operators must decide whether to start with a service request for emergency or to deploy emergency-call-over-NR directly.

SMS continues to be an important service in the 5GS. There are two basic methods to transport SMS in the combined 5GS and EPS, namely SMS-over-IP (SMSoIP) using an IMS SIP (Session Initiation Protocol) message and SMS-over-NAS (SMSoNAS) using NAS signaling. The latter uses 5G NAS when the device is in the 5GS and 4G NAS when in the EPS. Operators that use SMSoNAS for devices in the EPS are migrating to support SMSoNAS in the 5GS. Operators that are already using SMSoIP in the EPS may continue to support SMSoIP over the 5GS.

Beyond MBB, 5G also includes a multitude of new and enhanced capabilities addressing many different business segments. It is difficult to predict which of these capabilities will be required in the early introduction of the 5GS and the answer may vary for different operators. Examples of frequently mentioned opportunities for wide-area services are automotive [5], smart grids for utilities, and public safety.

All of these use cases will utilize the established 5GS MBB scale and wide-area network build-out, specifically if some key architectural concepts are planned for when introducing the 5GS. These include technologies like network slicing, edge computing and network exposure [6], for which the required functionality is already built-in from the start. These 5GS technologies are also key enablers for local network deployments at hospitals, harbors, airports and manufacturing facilities [7].

…5GS TECHNOLOGIES ARE ALSO KEY ENABLERS FOR LOCAL NETWORK DEPLOYMENTS…

Conclusion

Introducing the 5G System for wide-area services in an existing Evolved Packet System network has significant impacts across all network domains, including the RAN, packet core, user data and policies, and services, as well as affecting devices and backend systems. Nonetheless, a combined 4G-5G network is a necessary step for most operators with existing EPS deployments.

MIGRATION FROM EPS TO 5GS ✱

Successful management of this transition requires a holistic strategy that considers all network domains, as well as coverage strategy, spectrum assets and which services to offer where. There are several supported migration paths per domain to a full 5GS, and the transition can be adapted to address each operator's specific needs per domain. In the longer term, the introduction of 5GS supporting mobile broadband as initial wide-area service is a solid foundation to introduce additional services and use cases, meeting the full expectation on a future-proof 5GS.

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Further reading

- **5G is here, available at:** *https://www.ericsson.com/en/5g*
- **5G Voice, available at:** *https://www.ericsson.com/en/digital-services/trending/5g-voice-evolution-where-to-start*
- **5G Access, available at:** *https://www.ericsson.com/en/networks/offerings/5g*
- **5G Core, available at:** *https://www.ericsson.com/en/digital-services/offerings/core-network/5g-core*

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5G evolution:

3GPP RELEASES 16 & 17 OVERVIEW

The enhancements in the 3GPP releases 16 and 17 of 5G New Radio include both extensions to existing features as well as features that address new verticals and deployment scenarios. Operation in unlicensed spectrum, intelligent transportation systems, Industrial Internet of Things, and non-terrestrial networks are just a few of the highlights.

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According to the latest Ericsson Mobility Report, global traffic levels hit 38 exabytes per month at the end of 2019, with a projected fourfold increase to 160 exabytes per month expected by 2025 [1]. Fortunately, the 5G system is designed to handle this massive increase in data traffic in a way that ensures superior performance with minimal impact on the net costs for consumers.

■ The evolution of 5G New Radio (NR) has progressed swiftly since the 3GPP standardized the first NR release (release 15) in mid-2018. Not only is release 16 nearly finalized but the scope of release 17 has also recently been approved. Making wise decisions in the months and years ahead will require that mobile network operators and other industry stakeholders have a solid understanding of both releases.

NR development started in release 15 [2] [3] with the ambition to fulfill the 5G requirements set by the ITU (International Telecommunication Union) in IMT-2020 (International Mobile Telecommunications-2020). The overall design consists of several key components. The extension to much higher carrier frequencies is an important one due to the continuing demand for more traffic and higher consumer data rates and the associated need for more spectrum and wider transmission bandwidths. The ultra-lean design of NR enhances network energy performance and reduces interference, while interworking and LTE coexistence will make it possible to utilize existing cellular networks. The forward compatibility of NR design will ensure that it is prepared for future evolution. Low latency is also critical to improve performance and enable new use cases. Extensive usage of beamforming and a massive number of antenna elements for data transmission and for control-plane procedures are also notable components of NR design.

Figure 1 shows the time plan for the evolution of NR over the next few years. Release 16, the first step in the NR evolution, contains several significant extensions and enhancements. Some of these are extensions/improvements to existing features, while others are entirely new features that address new deployment scenarios and/or new verticals.

5G NR release 16 – enhancements to existing features

The most notable enhancements to existing features in release 16 are in the areas of multiple-input, multiple-output (MIMO) and beamforming enhancements, dynamic spectrum sharing (DSS), dual connectivity (DC) and carrier aggregation (CA), and user equipment (UE) power saving.

Multiple-input, multiple-output and beamforming enhancements

Release 16 introduces enhanced beam handling and channel-state information (CSI) feedback, as well as support for transmission to a single UE from multiple transmission points (multi-TRP) and full-power transmission from multiple UE antennas in the uplink (UL). These enhancements increase throughput, reduce overhead, and/or provide additional robustness [4]. Additional mobility enhancements enable reduced handover delays, in particular when applied to beam-management mechanisms used for deployments in millimeter (mm) wave bands [5].

Dynamic spectrum sharing

DSS provides a cost-effective and efficient solution for enabling a smooth transition from 4G to 5G by allowing LTE and NR to share the same carrier. In release 16, the number of rate-matching patterns available in NR has been increased to allow spectrum sharing when CA is used for LTE.

Dual connectivity and carrier aggregation

Release 16 reduces latency for setup and activation of CA/DC, thereby leading to improved system capacity and the ability to achieve higher data rates. Unlike release 15, where measurement configuration and reporting does not take place until the UE enters the fully connected state, in release 16 the connection can be resumed after periods of inactivity without the need for extensive signaling for configuration and reporting [6]. Additionally, release 16 introduces aperiodic triggering of CSI reference signal transmissions in case of the aggregation of carriers with different numerology.

User equipment power saving

To reduce UE power consumption, release 16 includes a wake-up signal along with enhancements to control signaling and scheduling mechanisms [7].

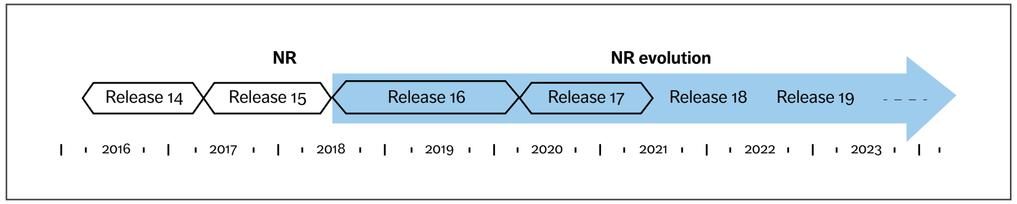

Figure 1 NR evolution time plan

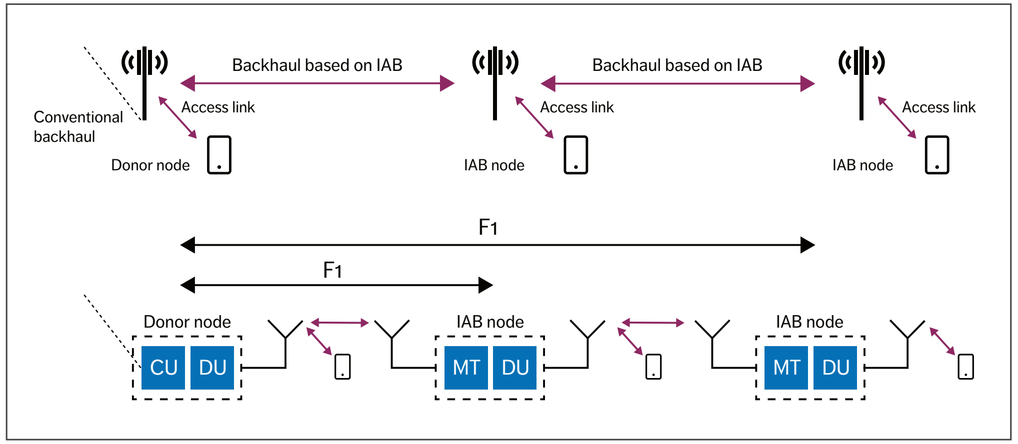

Figure 2 High-level architecture of IAB

5G NR release 16 – new verticals and deployment scenarios

The most notable new verticals and deployment scenarios addressed in release 16 are in the areas of:

- Integrated access and backhaul (IAB)
- NR in unlicensed spectrum
- Features related to Industrial Internet of Things (IIoT) and ultra-reliable low latency communication (URLLC)
- Intelligent transportation systems (ITS) and vehicle-to-anything (V2X) communications
- Positioning.

Integrated access and backhauling

IAB provides an alternative to fiber backhaul by extending NR to support wireless backhaul [8]. As a result, it is possible to use NR for a wireless link from central locations to distributed cell sites and between cell sites. This can simplify the deployment of small cells, for example, and be useful for temporary deployments for special events or emergency situations. IAB can be used in any frequency band in which NR can operate. However, it is anticipated that mm-wave spectrum will be the most relevant spectrum for the backhaul link. Furthermore, the access link may either operate in the same frequency band as the backhaul link (known as inband operation) or by using a separate frequency band (out-of-band operation).

Architecture-wise, IAB is based on the CU/DU split introduced in release 15. The CU/DU split implies that the base station is split into two parts – a centralized unit (CU) and one or more distributed units (DUs) – where the CU and DU(s) may be physically separated depending on the deployment. The CU includes the RRC (radio resource control) and PDC (packet data convergence) protocols, while the DU includes the RLC (radio link control) and MAC (multiple access control) protocols along with the physical layer. The CU and DU are connected through the standardized F1 interface.

Figure 2 illustrates the basic structure of a network utilizing IAB. The IAB node creates cells of its own and appears as a normal base station to UEs connecting to it. Connecting the IAB node to the network uses the same initial-access mechanism as a terminal. Once connected, the IAB node receives the necessary configuration from the donor node. Additional IAB nodes can connect to the network through the cells created by an IAB node, thereby enabling multi-hop wireless backhauling.

The lower part of the figure highlights that an IAB node includes a conventional DU part that creates cells to which UEs and other IAB nodes can connect. The IAB node also includes a mobile-termination (MT) part providing connectivity for the IAB node to (the DU of) the donor node.

New Radio in unlicensed spectrum

Spectrum availability is essential to wireless communication, and the large amount of spectrum available in unlicensed bands is attractive for increasing data rates and capacity for 3GPP systems. To exploit this spectrum resource, release 16 enables NR operation in unlicensed spectrum, targeting the 5GHz and 6GHz unlicensed bands. It supports both standalone operation, where no licensed spectrum is necessary, and licensed-assisted operation, where a carrier in licensed spectrum aids the connection setup. This greatly adds to deployment flexibility compared with LTE, where only licensed-assisted operation is supported.

Operation in unlicensed spectrum is dependent on several key principles including ultra-lean transmission and use of the flexible NR frame structure. Both of these were included in release 15. Channel access mechanisms based on listen-before-talk (LBT) are probably the most obvious area of enhancement in release 16. NR largely reuses the same LBT mechanism as defined for Wi-Fi and LTE in unlicensed spectrum. Interestingly, it was demonstrated during standardization that replacing one Wi-Fi network with an NR network can lead to improved performance for the remaining Wi-Fi networks [9] as well as for the NR network itself.

Industrial IoT and ultra-reliable low-latency communication

The IIOT is a major vertical focus area for NR release 16. To widen the set of potential IIoT use cases and support increased demand for new use cases such as factory automation, electrical power distribution and the transport industry, release 16 includes latency and reliability enhancements that build on the already very low air-interface latency and high reliability [10] provided by release 15. Support for time-sensitive networking (TSN), where very accurate time synchronization is essential, is also introduced. *Figure 3* illustrates TSN integration in 5G NR.

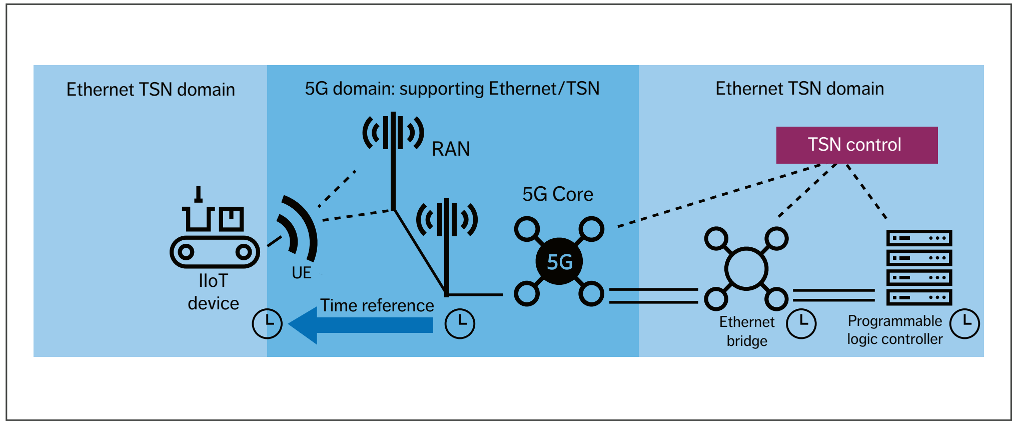

Figure 3 Overview of the TSN integration

Although many of the URLLC-related improvements are small in themselves, taken together they significantly enhance NR in the area of URLLC [11].

The inter-UE downlink (DL) preemption that is already supported in release 15 is extended in release 16 to include the UL, such that a UE's previously scheduled lower-priority UL transmission can be preempted (that is, cancelled) by another UE's higher-priority UL transmission. Release 16 also supports standardized handling of intra-EU UL resource conflicts.

To reduce latency, release 16 supports more frequent control-channel monitoring. Furthermore, for both UL configured grant and DL semi-persistent scheduling, multiple configurations can be active simultaneously to support multiple services. These enhancements are especially useful in combination with TSN traffic, where the traffic pattern is known to the base station.

CURRENTLY, 25 USE CASES FOR ADVANCED V2X COMMUNICATIONS HAVE BEEN DEFINED

Intelligent transportation systems and vehicle-to-anything communications

ITS, which provide a range of transport and traffic-management services, are another major vertical focus area in release 16. Among other benefits, ITS solutions improve traffic safety as well as reducing traffic congestion, fuel consumption and environmental impacts. To facilitate ITS, communication is required not only between vehicles and the fixed infrastructure but also between vehicles. Currently, 25 use cases for advanced V2X communications have been defined, including vehicle platooning and cooperative communication using extended sensors [12].

In release 15, communication with fixed infrastructure is provided by the access-link interface between the base station and the UE. Release 16 adds the option of the NR sidelink (PC5), which can operate in in-coverage, out-of-coverage and partial-coverage scenarios, utilizing all NR frequency bands. It supports unicast, groupcast and broadcast communication, and hybrid automatic repeat request (hybrid-ARQ) retransmissions can be used for scenarios that require more robust communication. Groups can be either configured or formed, and the group members communicate using groupcast transmissions. A truck platoon, for example, could be configured using dedicated hybrid-ARQ signaling between the receivers and transmitter, or formed in a dynamic manner based on the distance between the transmitter and receiver(s).

Positioning

For many years, UE positioning has been accomplished with Global Navigation Satellite Systems assisted by cellular networks. This approach provides accurate positioning but is typically limited to outdoor areas with satellite visibility. There is currently a range of applications that requires accurate positioning not only outdoors but also indoors. Architecture-wise, NR positioning is based on the use of a location server, similar to LTE. The location server collects and distributes information related to positioning (UE capabilities, assistance data, measurements, position estimates and so on) to the other entities involved in the positioning procedures. A range of positioning methods, both DL-based and UL-based, are used separately or in combination to meet the accuracy requirements for different scenarios.

DL-based positioning is supported by providing a new reference signal called the positioning reference signal (PRS). Compared with LTE, the PRS has a more regular structure and a much larger bandwidth, which allows for a more precise correlation and time of arrival (ToA) estimation. The UE can then report the ToA difference for PRSs received from multiple distinct base stations, and the location server can use the reports to determine the position of the UE.

THE ENHANCEMENTS... RELATE TO SPECIFIC NEW REQUIREMENTS THAT ARE EMERGING IN THE MARKET

UL-based positioning is based on release 15 sounding reference signals (SRSs) with release 16 extensions. Based on the received SRSs, the base stations can measure and report (to the location server) the arrival time, the received power and the angle of arrival from which the position of the UE can be estimated.

The time difference between DL reception and UL transmission can also be reported and used in round-trip time (RTT) based positioning schemes, where the distance between a base station and a UE can be determined based on the estimated RTT. By combining several such RTT measurements, involving different base stations, the position can be determined.

5G NR release 17

The work items approved by the 3GPP in December 2019 will lead to the introduction of new features for the three main use case families: enhanced mobile broadband (eMBB), URLLC and massive machine-type communications (mMTC). The purpose is to support the expected growth in mobile-data traffic, as well as customizing NR for automotive, logistics, public safety, media and manufacturing use cases. The enhancements to existing features introduced in release 17 will be for functionality already deployed in live NR networks or relate to specific new requirements that are emerging in the market. The table presented in *Table 1* summarizes the scope of the enhancements to existing NR features in release 17, while the table in *Table 2* summarizes the scope of the new features.

Terms and abbreviations

CA – Carrier Aggregation | **CSI** – Channel-State Information | **CU** – Centralized Unit | **DC** – Dual Connectivity | **DL** – Downlink | **DSS** – Dynamic Spectrum Sharing | **DU** – Distributed Unit | **eMBB** – Enhanced Mobile Broadband | **FR1, FR2** – Frequency Range 1, 2 | **hyrbrid-ARQ** – Hybrid Automatic Repeat Request | **IAB** – Integrated Access and Backhaul/Backhauling | **IIoT** – Industrial Internet of Things | **IoT** – Internet of Things | **ITS** – Intelligent Transportation Systems | **LBT** – Listen-Before-Talk | **LTE-M** – LTE Machine-Type Communications | **MIMO** – Multiple-Input, Multiple-Output | **mMTC** – Massive Machine-Type Communications | **MT** – Mobile Termination | **MTC** – Machine-Type Communications | **Multi-TRP** – Multiple Transmission Points | **NR** – New Radio | **PC5** – Direct Mode Interface | **PRS** – Positioning Reference Signal | **RTT** – Round-Trip Time | **SON** – Self-Organizing Networks | **SRS** – Sounding Reference Signal | **ToA** – Time of Arrival | **TSN** – Time-Sensitive Networking | **UE** – User Equipment | **UL** – Uplink | **URLLC** – Ultra-Reliable Low-Latency Communication | **V2X** – Vehicle-to-Anything | **XR** – Anything Reality

eMBB feature	
IAB	• Addition of (limited) support for network topology changes • Improved duplexing of access and backhaul links (simultaneous operation on child and parent link, for example) • Routing enhancements
MIMO	• Improvements based on experience from commercial networks focusing on multi-beam operation mainly for frequency range 2 (FR2), support for multi-TRP deployment, SRSs, and CSI measurement and reporting
DSS	• Cross-carrier scheduling enhancements • Other scheduling enhancements
Coverage	• Enhanced wide-area coverage for both FR1 and FR2 (to be studied) • Focus on mobile broadband and voice services use cases, with the exception of the low-power wide area use case
Multi-radio dual connectivity	• More efficient activation/deactivation mechanism of secondary cells • Conditional primary-secondary cell change/addition
UE power saving	• Improved mechanisms in the area of discontinuous reception and blind decoding of control channels
Data collection	• Simplified deployment and enhancements to support self-organizing networks (SON) with improved data-collection mechanisms for SON and minimization of drive tests
QoE management and optimizations for diverse services	• Generic framework for triggering and configuring QoE measurement collection and reporting for various 5G use cases
URLLC feature	
IIoT and URLLC support	• Improved support for factory automation and URLLC, including physical layer feedback enhancements and enhancements for support of time synchronization • Identification of enhancements for URLLC/IIoT operation in controlled environments on unlicensed bands
Positioning	• Higher accuracy (horizontal and vertical) and lower latency, especially for IIoT use cases
Sidelink	• Focus on V2X, public safety and commercial use cases • Resource allocation enhancement • Sidelink discontinuous reception
RAN slicing (also relevant for the mMTC use case)	• Mechanisms to enable UE fast access to the cell supporting the intended slice • Mechanisms to support service continuity for intra-radio-access technology handover service interruption
mMTC feature	
Small data transmissions in inactive state	• Reduced overhead from connection establishment • Use cases: keep-alive messages, wearables and various sensors

Table 1 Summary of release 17 enhancements to existing features

eMBB feature	
Supporting NR from 52.6GHz to 71GHz	• Extended NR frequency range to allow exploitation of more spectrum, including the 60GHz unlicensed band • Definition of new OFDM (orthogonal frequency-division multiplexing) numerology and channel access mechanism to comply with the regulatory requirements applicable to unlicensed spectrum
Multicast and broadcast services	• Primarily targeted at V2X, public safety, IP multicast, software delivery and Internet of Things (IoT) applications
Support for multi-SIM devices	• Paging collision avoidance • Network notification when a UE switches networks
Support for non-terrestrial networks	• Support for satellites (especially Low Earth orbit and geostationary satellites) and high-altitude platforms as an additional means to provide coverage in rural areas
Multi-radio dual connectivity	• L2 versus L3 relaying (study and compare) • Scenarios include single-hop, UE-to-UE and UE-to-network relaying
Sidelink relaying	• Improved mechanisms in the area of discontinuous reception and blind decoding of control channels
Data collection	• Simplified deployment and enhancements to support SON with improved data-collection mechanisms for SON and minimization of drive tests
URLLC feature	
Anything reality (XR) evaluations	• Evaluate needs in terms of simultaneously providing very high data rates and low latency in a resource-efficient manner • Intended to support various forms of augmented reality and virtual reality, collectively referred to as XR
mMTC feature	
Support of reduced-capability NR devices	• Targeted at mid-tier applications such as machine-type communications for industrial sensors, video surveillance, and wearables with data rates between Narrowband IoT/LTE-M data rates and “full” NR data rates • Addresses issues including complexity reduction, UE power saving and battery lifetime enhancement

Table 2 Summary of new functionality added in release 17

5G NR EVOLUTION ✱

Conclusion

The enhancements in the 3GPP's releases 16 and 17 will play a critical role in expanding both the availability and the applicability of 5G New Radio to a wide range of new applications and use cases in both industry and public services. In order to make the details of these two releases more easily digestible, we have identified what we consider to be the most significant enhancements and grouped them into two categories: enhancements to existing features and features that address new verticals and deployment scenarios.

From Ericsson's point of view, the overall ambition of the NR evolution from a use-case perspective must be to ensure that 5G NR covers all relevant use cases to fulfill the vision of ubiquitous connectivity – that is, the ability to connect anything anywhere at any time. From a features perspective, we believe that the evolution of NR functionality must be driven by the goal of increasing efficiency and effectiveness when and where it is commercially justified.

Looking ahead, it is critical that the industry works together to ensure that NR is easy to deploy and operate, and that it continues to provide superior performance compared with competing technologies. We must also ensure that NR provides a high degree of energy efficiency on both the network and device sides, and that it retains its ability to coexist smoothly with LTE.

At Ericsson, we are convinced that the best way forward is for NR to continue to support all use cases from one platform, with a focus on forward compatibility, sufficient configurability and maximal simplicity. We must also work to avoid unnecessary updates in the network hardware and ensure that functionality is specified in a common way that benefits multiple use cases.

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Further reading

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- **A new standard for Dynamic Spectrum Sharing, available at:** *https://www.ericsson.com/en/blog/2019/6/dynamic-spectrum-sharing-standardization*
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